

WHEN CAN WOMEN CLOSE THE GAP? A META-ANALYTIC TEST OF SEX DIFFERENCES IN PERFORMANCE AND REWARDS

APARNA JOSHI
Pennsylvania State University

JOOYEON SON
University of Illinois, Urbana-Champaign

HYUNTAK ROH
Yonsei University

Drawing on macro and micro domains in gender research, we meta-analytically test whether occupation-, industry-, and job-level factors mitigate or exacerbate differences in performance evaluations ($k = 93$; $n = 95,882$) and rewards ($k = 97$; $n = 378,850$) between men and women. Based on studies conducted across a variety of work settings and spanning nearly 30 years, we found that the sex differences in rewards ($d = .56$) (including salary, bonuses, and promotions) were 14 times larger than sex differences in performance evaluations ($d = .04$), and that differences in performance evaluations did not explain reward differences between men and women. The percentage of men in an occupation and the complexity of jobs performed by employees enhanced the male–female gap in performance and rewards. In highly prestigious occupations, women performed equally, but were rewarded significantly lower than men. Only a higher representation of female executives at the industry level enabled women to reverse the gender gap in rewards and performance evaluations. Our configurational analysis also revealed that some occupation-, industry-, and job-level attributes of the work context jointly contributed to differences in rewards and performance evaluations.

In 1982, when Ann Hopkins was denied partnership at a prestigious accounting firm despite a stellar performance record, her attorneys presented compelling evidence before the Supreme Court of the United States that the decision was a direct outcome of gender-based stereotyping and discrimination in the firm (Fiske, Bersoff, Borgida, Deaux, & Heilman, 1991; see also *Price Waterhouse v. Hopkins* 109 S. Ct. 1775 (1989)). Such instances of overt stereotyping and egregious gender discrimination are rare in workplaces today. Nevertheless, more than two decades since the historic Supreme Court ruling in favor of Ann Hopkins, women continue to remain grossly underrepresented at the

highest levels in organizations, and in many settings receive significantly lower pay and fewer promotions than men (Catalyst, 2008). Indeed, scholars recognize that detecting or remedying gender-based stereotyping, bias, or discrimination continues to pose a substantial challenge in organizations (e.g., Blau & Kahn, 2007; Kalev, Dobbin, & Kelly, 2006; Reskin, 2003). Litigation pressures and the diffusion of symbolic diversity management practices obscure bias and stereotyping, making it unlikely that purposive discrimination attributable to specific organizational agents can be easily detected (Ortiz & Roscigno, 2009; Tilly, 1998). Therefore, rather than focusing research on detecting the (often unmeasurable) psychological processes that underlie gender discrimination at work, some scholars have noted that identifying objective (often measurable) attributes of the workplace context associated with sex differences in employment outcomes offers one important avenue for building an actionable theory of gender inequality (Baron & Pfeffer, 1994; Reskin, 2003).

We thank Jason Colquitt and three anonymous reviewers for their valuable feedback throughout the review process. We are also very grateful to Vilmos Misangyi and Daniel Newman for their helpful feedback on previous versions of the manuscript. Funding for this research was provided by a National Science Foundation Grant (#1307353).

As a step in this direction, we develop and test a theoretical framework that draws on the cumulative research evidence, spanning nearly three decades, on sex differences in organizational rewards and performance evaluations. This evidence points out that although sex differences in organizational rewards and performance evaluations are fairly widespread, the gap between women and men also varies considerably across work contexts. Certainly, in many settings, men outperform women and receive higher rewards, such as pay and promotions (e.g., Greenhaus & Parasuraman, 1993; Sackett, DuBois, & Noe, 1991). However, in some situations, women do receive more favorable performance evaluations and are also able to close the gap in organizational rewards (e.g., Dencker, 2008; Speilerman & Petersen, 1999). The varied nature of the gap between women and men across work contexts offers important directions for further inquiry; it suggests that some contexts may be more susceptible and others more resilient to gender inequality. Building on this premise, our approach seeks to develop a theoretical framework that identifies several structural attributes of the work context that can reinforce or mitigate differences in performance and reward outcomes between men and women.

This framework offers a substantive step forward in gender inequality research for a number of reasons. In both micro (e.g., Cleveland, Vescio, & Barnes-Farrell, 2005) and macro (e.g., DiTomaso, Post, Parks-Yancy, 2007) domains of gender research, there have been calls to integrate across disciplinary silos to better understand the prevalence of inequity at work. Indeed, more than two decades ago, Baron and Pfeffer (1994: 191) noted that:

... missing in most literature on (inequity in organizations) is any attention to the micro-macro connection, the links between social structures, institutions, and organizations on the one hand and, on the other, cognitions, perceptions, interests and behaviors at the individual or small-group level.

To date, management research, although uniquely positioned to make this “micro-macro” connection, has not fully responded to these calls.

We aim at developing a framework that bridges the micro-macro divide in sex differences research by uniting institutional and structural perspectives on gender inequality (e.g., Stainback, Tomaskovic-Devey, & Skaggs, 2010) with sociopsychological perspectives on gender bias (e.g., Eagly, Makhijani, & Klonsky, 1992), to specify how aspects of the work context predict sex differences in performance evaluations

and organizational rewards. Based on these theoretical perspectives, we meta-analytically test the effects of occupation-, industry-, and job-level predictors on sex differences in performance evaluations and reward outcomes among men and women performing comparable jobs. For example, our approach allows us to examine whether the level of complexity of jobs performed by investment bankers explains why female investment bankers earn significantly lower bonuses than male investment bankers, or whether greater parity in performance evaluations among male and female software developers can be attributed to a high proportion of female executives in the information technology (IT) industry. Drawing on multiple strands in gender research, we propose that although gender bias or stereotyping can potentially operate in each of these contexts, factors such as job complexity or representation of female executives at the industry level can strengthen or weaken the extent to which these processes determine employment outcomes for men and women. Based on this framework, we recommend directions for future research on the role that performance evaluation and reward allocation decision making can play in shaping sex differences in rewards or performance within specific contexts.

A second important contribution of this research is that we examine the effects of contextual attributes on both performance evaluations and reward differences between men and women. In order to unpack the sources of gender inequality, it is important to understand whether or not contextual factors that mitigate or enhance sex differences in performance also mitigate or enhance sex differences in pay and promotions (Blau & Kahn, 2007; Cleveland et al., 2005). It is often assumed that psychological processes or structural attributes that drive sex differences in performance evaluations also drive sex differences in outcomes such as pay, promotions, or other organizational rewards. We propose that this assumption needs to be examined more carefully. Consider that, in many contexts, reward allocation decisions (e.g., promotion to equity partnership) often imply zero-sum choices among a pool of qualified candidates, while performance criteria (e.g., billable hours) are not subject to such considerations. By comparing sex differences in performance evaluations versus rewards across various work settings, we hope to shed light on whether various contextual variables have similar effects on these outcomes. Put simply, our framework allows us to test whether in contexts in which men are rewarded at higher levels than women, they also receive higher performance evaluations

than women, and in contexts in which women receive higher evaluations, they also receive higher rewards than men.

On a related note, we also directly test whether performance differences *explain* reward differences among men and women. Research that has examined differences in pay and promotions among men and women rarely account for performance differences (see Castilla, 2008; Blau & Devaro, 2007, as exceptions). Instead, it is often assumed that productivity differences, measured as educational attainment or labor market experience, drive sex differences in rewards (Tomaskovic-Devey, 1993). However, the educational and skill attainments that women have made since the 1980s have surpassed even average workforce-level human capital gains during the period (Bernhardt, Morris, & Handcock, 1995; Blau & Kahn, 2007). These gains should enable women to close any performance gaps with their male counterparts, but the question remains: Do these gains also allow them to close the gender gap in organizational rewards? Our framework integrates past research on sex differences in pay with research on sex differences in performance evaluations to answer this question.

Our meta-analysis is also unique in that we apply a configurational approach to supplement meta-regression procedures and to identify whether *bundles* of occupational, industry, and job attributes jointly contribute to sex differences in performance evaluation and organizational rewards (Fiss, 2011; Ragin, 2008). This approach allows us to identify which contextual factors combine, complement, or act as substitutes for each other in situations in which men outperform, or are rewarded significantly higher than, women. Such an approach also enables us to highlight less obvious combinations of contextual attributes that may be sufficient for observing sex differences in performance evaluations and rewards.

MACRO-MICRO LINKAGES IN GENDER RESEARCH

The proposed framework tests the notion that sex differences in organizational rewards and performance evaluations are a function of “what” the nature of the work is, as well as “where” the work is being done (Reskin, 2003). Reskin (2003: 14) notes that these aspects of “structure and context are fundamental concepts because they highlight the importance of [the] setting on social processes that govern [gender] inequality in the workplace.” Developing this logic, we focus on how job-level (i.e., what the

nature of the work is), as well as occupation- and industry-level (i.e., where the work is being done), attributes can shape sex differences in performance evaluations and rewards in an organization.

Consider that any given work organization operates within a larger industry environment and employs individuals across a diverse array of occupations, performing varied tasks ranging from complex to routine. Recognizing that such an organization (particularly in the United States) also has considerable discretion in designing performance management and compensation practices (Nelson & Bridges, 1999), we propose that specific job, industry, and occupational factors define the overall environment in which organizational agents such as human resources (HR) professionals, executives, and supervisors make evaluative and resource allocation decisions (Dobbin, 2009; Tilly, 1998). This environment comprises specific cultural norms, stereotypic expectations, and status cues associated with gender. This environment also reflects the institutional or market forces that define a work context and influence employment outcomes for men and women. Within this embedding context, we discuss below how reward allocation and performance evaluation practices advantaging men over women can potentially take on an “everyday,” or “business as usual,” quality (Ely & Padavic, 2007; Hultin & Szulkin, 2003; Tilly, 1998).

Occupational Effects on Sex Differences in Performance Evaluations and Rewards

In the past, some research accounts linking occupations to gender inequality have emphasized women’s skill levels and related job choices as antecedents of women’s segregation into lower paying occupations (Bergmann, 1974; Blau & Kahn, 1981). But this research does not fully explain why gender inequality is prevalent despite women’s human capital gains and a steady entry into traditionally male-dominated or higher paying occupations (Gatta & Roos, 2005). We propose that overall prestige and gender typing of an occupation defines normative role expectations, status cues, and stereotypic beliefs about how occupational incumbents are perceived, evaluated, and rewarded, providing important additional explanations for persistent differences between men and women.

Occupational demographic composition. The demographic make-up of an occupation can signal the “appropriateness,” or “fit,” of an occupation for men and for women, driving stereotypic expectations

of role and abilities associated with that occupation (Eagly & Makhijani, 2002; Gorman, 2005; Heilman, 1983). Occupations that are female-dominated (such as nursing or elementary school teaching) may be considered a more appropriate “fit” for women, while occupations that are male-dominated (such as fire-fighting or production engineering) may be considered appropriate for men.

When women enter highly male-dominated occupations, they do not “fit” the stereotypic expectations of abilities expected in that occupation, and therefore experience greater bias and discrimination (Eagly & Makhijani, 2002; Heilman, 1983). Role congruity theory, which has been primarily applied to women in leadership roles, posits that even when female leaders display high levels of performance, their efforts may be discounted by their peers and/or supervisors (e.g., Eagly et al., 1992). This theory recognizes that culturally shared beliefs about the appropriate roles and abilities of men and women in society have widespread effects in the workplace (Eagly & Karau, 1991; Eagly et al., 1992). As such, the contributions of women who occupy roles or display abilities that are atypical relative to established cultural norms tend to be undervalued and discounted at work (Eagly & Karau, 1991; Eagly et al., 1992; Kanter, 1977). These gender-based attributions of skills are often not essential, or even beneficial, for performing various aspects of a job, but can still influence how work is defined and valued in organizations (Ely & Padavic, 2007).

Returning to our opening example, a factor contributing to gender bias in Ann Hopkins’ performance appraisals was her lack of fit in the broad occupational category of professional accountants—an occupational category dominated by men (Fiske et al., 1991). If Ann Hopkins had been an administrative assistant (i.e., in a female-dominated occupational category) in the firm, it is less likely that she would have faced the types of barrier that she encountered. Experimental research shows that women in male-typed settings encounter bias at two levels: First, women are viewed as having less male-typed ability than men (e.g., women are generally viewed as having less agentic leadership ability than men); and second, women who do display male-typed behavior are viewed as less effective than men who display the same behavior (Eagly & Karau, 1991). More recent extensions of this theory have also identified a backlash against female leaders who are perceived as threatening to the male-dominant status quo (Rudman, Moss-Racusin, Phelan, & Nauts, 2012). These perspectives suggest that, in male-dominated occupations, women

may face a “double penalty” when their performance is evaluated, or they are considered for promotions or for bonuses. First, supervisors may view women as less qualified to perform a job or to receive a promotion because their skills or abilities do not fit male-typed norms and expectations. Second, when women do display male-typed behaviors that run counter to female-typed stereotypic expectations, they may be viewed as threatening to the status quo and may be evaluated negatively or denied opportunities for career advancement (Fiske et al., 1991; Rudman et al., 2012).

Across many male-typed occupational contexts, research provides general support for this line of argument. Based on survey data from scientists and engineers across U.S. corporations, research shows that, in these traditionally male-typed occupations, white men received significantly more access to skill development opportunities and higher performance evaluations than any other demographic group, including white women, women of color, or immigrant men (DiTomaso, Post, Smith, Farris & Cordero, 2007). In the context of science and engineering teams, based on “round robin” data gathered from team members, Joshi (2014) found that gender served as an expertise cue such that male scientists received higher expertise evaluations from their male and female colleagues, and, in some circumstances, highly educated women were evaluated *lower* than less educated women by male team members. Men also received more opportunities than women to collaborate in ongoing research projects within teams (Joshi, 2014).

Across other occupational contexts as well, research shows that membership in a demographic group that is dominant in an occupation predicts access to important information such as the availability of training programs and job opportunities (Tomaskovic-Devey & Skaggs, 1999). Cumulatively, this research suggests that the demographic composition of the occupation—specifically, the predominance of men—reflects a cultural context (i.e., gender-related norms, stereotypes, and status cues) that shapes reward allocation and evaluation behavior, with potentially negative employment consequences for women. Thus we posit that:

Hypothesis 1a. The proportion of men in an occupational category enhances sex differences in performance evaluations such that men receive higher performance evaluations than women in occupations with a higher proportion of men.

Hypothesis 1b. The proportion of men in an occupational category enhances sex differences in organizational rewards such that men receive higher rewards than women in occupations with a higher proportion of men.

Occupational prestige. Occupational prestige, the socioeconomic value of an occupation, has been acknowledged as an important variable predicting employment outcomes because it foreshadows occupational mobility and is a proxy for social class (Hodge, Siegel, & Rossi, 1964). The prestige of an occupation may be related to, but also operates independently from, the sex composition of the occupation in predicting income differences between men and women (England, 1979; England & McLaughlin, 1979; Siegel, 1971). Indeed, some highly prestigious occupations, such as dentist or psychologist, are fairly gender-balanced, whereas others, such as astronaut or physician, tend to be male-dominated. Some occupations, such as “automotive salesperson” or “warehouse worker,” are male-dominated, but less prestigious. Not only are occupational prestige hierarchies extremely stable across time, but also there is widespread consensus within societies regarding which occupations are more or less prestigious (Hodge et al., 1964). Because highly prestigious occupations, representing higher social class, require higher investments in human capital (consider lawyers or physicians), we might expect the men and women working in these settings to be similarly qualified and not to experience differential employment outcomes. However, women face several barriers in these contexts as well.

In highly prestigious occupations, barriers to entry and advancement are extremely high. Access to advancement in these settings is controlled via mechanisms such as licensing, formal educational requirements, certification, and sometimes opaque performance and reward criteria (e.g., Weeden, 2002). Research based on social dominance theory suggest that formal and informal practices in these occupational settings constitute a hierarchy-enhancing context (Sidanius & Pratto, 1999; Sidanius, Pratto, Sinclair & van Laar, 1996; Sinclair, Sidanius, & Levin, 2010). “Hierarchy-enhancing agents”—a term broadly applied to social institutions or social ideologies—tend to “support and facilitate the disproportionate allocation of things with positive social value (e.g., high income, prestigious jobs, good education) to dominant social groups and things with negative social value (e.g., prison sentences, poor jobs, inferior education) to subordinate social groups”

(Sidanius, et al., 1996, p. 146). “Hierarchy-attenuating agents,” on the other hand, facilitate an egalitarian allocation of positively valued goods to dominant and subordinate social groups (Sidanius et al., 1996)

Highly prestigious occupations can operate as hierarchy-enhancing agents that favor the distribution of valued goods, such as positive performance appraisals or organizational rewards, to dominant social groups, such as men, over subordinate groups, such as women (Sinclair et al., 2010). Some research shows that, in these types of workplace, key decision makers involved in performance evaluations or wage-setting processes (senior executives, HR professionals, supervisors) are also likely to be high in social dominance orientation—a dispositional attribute that underlies a tendency to favor dominant over subordinate groups (Pratto, Stallworth, Sidanius, Siers, 1997). These dispositional tendencies of key decision makers and the prestige value of the context could serve as mutually reinforcing hierarchy-enhancing factors contributing to inequitable outcomes for women. Moreover, feminist scholars who have commented extensively on the intersections between gender and class note that gender and class relations operate jointly in these employment contexts (Acker, 2006; McCall, 2001). These scholars have argued that wage-setting and supervisory practices function as instruments of control over valued organizational resources, and serve to maintain the dominance of men in high-prestige work contexts that are also avenues for upward class mobility (e.g., Acker, 2006).

In these contexts, even seemingly meritocratic reward allocation and performance evaluation practices can function as hierarchy-enhancing agents by awarding more favorable outcomes to men than to women. For instance, in many prestigious settings, “up or out” promotion policies drive adversarial processes that are primed toward highlighting employees’ weaknesses. In these situations, evaluative decisions may draw on stereotypic information to highlight shortcomings (e.g., “not forceful enough,” “too emotional,” “lacks leadership qualities”) that tend to disadvantage women (Cleveland et al., 2005; Vescio, Gervais, Snyder, & Hoover, 2005). Across prestigious law firms, for example, recent research shows that senior partners making promotion decisions tended to value different types of client strategy across male and female associates: Women were more likely to be promoted when they brought in external clients, but men were promoted when they collaborated with senior partners within the firm (Briscoe & von Nordenflycht, 2014).

Although, on the face of it, both client strategies—internal and external—may contribute to firm revenues, bringing in external clients could be more

challenging and risky, imposing a higher burden on incoming female associates. Perhaps as a reflection of these subtle barriers, several studies in the context of law firms report that promotion rates are significantly lower and turnover rates significantly higher among women than among men (Gorman, 2006; Kay & Hagan, 1998; Spurr & Sueyoshi, 1994). Overall, these findings illustrate how subtle biases might permeate ostensibly meritocratic and gender-neutral performance evaluations and reward allocation practices in prestigious contexts, advantaging men over women (Martin & Myerson, 1998). In less prestigious contexts, barriers to entry and advancement are low, as are the stakes for controlling resources and maintaining the dominant status quo. In these occupational settings, the drive to reward dominant social groups over subordinate groups by means of performance evaluation and reward allocation practices is also likely to be lower, resulting in more equitable outcomes for women (Sinclair et al., 2010). Building on these insights, we propose that:

Hypothesis 2a. The prestige associated with an occupational category enhances sex differences in performance evaluations such that, in highly prestigious occupations, men receive higher performance evaluations than women.

Hypothesis 2b. The prestige associated with an occupational category enhances sex differences in organizational rewards such that, in highly prestigious occupations, men receive higher rewards than women.

Industry Effects on Sex Differences in Performance Evaluations and Rewards

The industry environment is also a distinct and relevant context shaping gender inequality in organizations. The industry context in which a firm operates refers to aspects of the business environment such as the regulatory framework, number and size of competitors, and the level of growth and globalization. Strategy scholars therefore regularly take into account industry attributes such as capital intensity, dynamism, and munificence in predicting firm performance (e.g., McGahan & Porter, 1997; Porter, 1980). The industry setting serves as an embedding context for firms, because firms within an industry compete within similar labor markets and face common institutional pressures that lead to the diffusion of employment practices across these firms (Stainback et al., 2010). Within a single firm encompassing multiple occupational categories, these institutional pressures could potentially influence the

adoption of reward allocation and performance evaluation practices underlying gender inequality.

The institutional context not only frames the adoption and diffusion of employment practices, but also can shape how employees respond to these practices. Research shows that, within an industry, perceptions of discrimination are socially constructed via regulatory and competitive pressures (Hirsch & Kornrich, 2008). Industrial categories are associated with broadly defined norms and expectations regarding antidiscrimination laws (Edelman, Fuller, & Mara-Drita, 2001). In fact, some scholars have found significant between-industry variations in the extent to which employees can “name” a practice as discriminatory, “blame” the employer for that practice, and “claim” redress for the discrimination (see Felstiner, Abel, & Sarat, 1980: 635–636.). Recognizing the importance of the broad industry setting, we propose more specifically that the ascendance of women to managerial and executive levels reflects institutional, labor market, and cultural pressures within an industry, and is an important contingency shaping sex differences in performance evaluations and rewards.

Research finds that the proportion of women in higher ranks of an organization serves as a significant constraint on gender inequality (e.g., Cohen & Broschak, 2013; Cohen, Broschak, & Haveman, 1998; Ely, 1994; Huffman, Cohen, & Pearlman, 2010). Women's representation in executive positions drives positive outcomes for junior women (Ely, 1994; 1995); senior women also pave the way for greater access to organizational power structures for other women (Huffman et al., 2010; Hultin & Szulkin, 2003). The demographic composition of managerial levels also has been found to reduce the gap in salary and in objective performance outcomes between men and women (Joshi, Liao, & Jackson, 2006; Ostroff & Atwater, 2003). A key logic underlying these findings is that the presence of women in positions of power in organizations has symbolic value—i.e., it signals greater status attainment among women. The presence of women at higher levels also provides more junior women access to career-related advice and mentoring (Ely, 1994; 1995). Finally, the presence of women at higher levels might also imply greater scrutiny of wage-setting, bonus allocation, and performance evaluation practices within organizations (Huffman et al., 2010; Joshi et al., 2006).

Although the research outlined above is focused at the organizational level, the growing role and visibility of industry-level professional women's networks (e.g., Women on Wall Street, the National Council for Women in Information Technology) suggests that

these arguments will also hold across firms within specific industry categories. These industry-wide network groups often organize conferences and workshops to expose women to visible symbols of success in the industry, and to help them to develop personal strategies to achieve career success. Thus the proportion of women in an industry can serve as a cultural context, shaping status cues and role expectations associated with gender. Moreover, within a specific industry category (consider healthcare versus automobile retailing), competitive, mimetic, or normative pressures to adopt more equitable employment practices will likely lead to greater representation of women at higher levels across firms (Dobbin, 2009; Stainback et al., 2010). Thus the representation of women in executive ranks within an industry is a reflection of greater scrutiny from regulatory bodies and employee groups; it is also a reflection of the diffusion of egalitarian employment practices in that industry. These attributes of the context should also facilitate equitable employment outcomes for men and women (Dobbin, 2009). Conversely, industries in which women have not been able to break into the upper echelons reflect contexts in which institutional pressures to adopt equitable practices have been less salient. In these settings, biased performance reviews or bonus and promotion decisions favoring men over women are more likely. Based on these arguments, we propose that:

Hypothesis 3a. The proportion of women at executive and senior managerial levels within a specific industry reduces sex differences in performance evaluations such that women receive equal to, or higher, performance evaluations than men in industries with a higher proportion of women at executive or managerial levels.

Hypothesis 3b. The proportion of women at executive and senior managerial levels within a specific industry reduces sex differences in organizational rewards such that women receive equal to, or higher, rewards than men in industries with a higher proportion of women at executive or managerial levels.

Job-Level Effects on Sex Differences in Performance Evaluations and Rewards

The job context encompasses a number of mechanisms by which male–female differences in employment outcomes manifest in organizations. For instance, the structure of jobs drives how incumbents

relate to each other and compare their employment circumstances. The structure of jobs also drives the definition and weighting of criteria for evaluating and rewarding incumbents (Lansberg, 1989). We focus on job complexity—i.e., the extent to which jobs involve a high level of problem-solving skills, discretion, and autonomy—as an important facet of the job context (Auster, 1989).

Research suggests that, in complex job settings, the cognitive demands placed on supervisors when evaluating performance and rewarding employees are fairly high; in these contexts, cognitive biases and stereotyping are more likely to play a role in supervisory performance evaluations and reward allocation (Auster, 1989; Fiske et al., 1991). In addition, discrimination and bias in evaluation is more likely to occur when information about evaluating a job cannot be easily collected and documented (Heilman, 1994; Nieva & Gutek, 1980). Some researchers note that employers may be more likely to behave stereotypically when there is limited, deniable, and ambiguous information regarding an individual's performance (Gerdes & Garber, 1983; Mobley, 1982). Complex jobs represent such settings. For instance, Mobley (1982) studied the performance appraisals of nonmanagement employees in a large supply organization, and found that evaluations of females in nonprofessional and nonmanagerial jobs are less likely to be affected by systematic gender bias owing to the lower complexity of nonmanagerial jobs.

Although the ambiguity or subjectivity of job performance criteria and the related propensity to stereotype against women is one important mechanism by which job complexity might enhance sex differences in performance evaluations, quantitative and narrative reviews suggest that subjective performance evaluation criteria alone do not always disadvantage women (Bartol, 1999; Roth, Purvis, & Bobko, 2012). We therefore posit that the ambiguity or subjectivity of the evaluation criteria may not be the only mechanism by which sex differences in performance evaluations or rewards emerge in complex jobs; rather, the complexity of the job context may shape subtler forms of bias that lead to systematic sex differences in these outcomes. For instance, complex jobs often involve more variable and demanding work schedules, as well as expectations that employees will put in “face time” after working hours. It is possible that although women are not necessarily disadvantaged in terms of performing the core tasks of a job, other intangible requirements of complex jobs might not work in their favor (Cleveland et al., 2005; Correll, Kelly, O'Connor, & Williams, 2014). Moreover, complex jobs are also

jobs with higher status and span of control in organizations (e.g., managerial and executive positions). Following the role congruence perspective outlined earlier (e.g., Eagly et al., 1992), it is also likely that women in these jobs will be considered atypical and face reward penalties.

Finally, job complexity is also associated with greater role autonomy, discretion, and idiosyncratic bargaining by incumbents, leading to an additional mechanism by which job complexity drives gender inequality: lack of opportunities for incumbents to compare objective employment conditions (Baron & Pfeffer, 1994). In complex jobs, because specific roles and terms and conditions of employment are less generalizable across incumbents, individuals are less likely to detect or report unequal treatment (Lansberg, 1989). Together, these perspectives suggest that job complexity is an important embedding context governing reward allocation and evaluative decisions in organizations. Thus we propose that:

Hypothesis 4a. Job complexity enhances sex differences in performance evaluations such that men receive higher performance evaluations than women in more complex jobs.

Hypothesis 4b. Job complexity enhances sex differences in organizational rewards such that men receive higher organizational rewards than women in more complex jobs.

Performance Differences as Explanations for Reward Differences between Men and Women

Although we have argued that sex differences in rewards and performance evaluations are primarily a function of “where” and “what” work is being performed, we also acknowledge that gender-based differences in performance evaluations may explain sex differences in organizational rewards. In field settings, studies that have accounted for performance differences in explaining reward differences are rare. Past research has shown that women receive fewer rewards in organizations even after accounting for attributes that might underlie performance differences, such as educational level and labor market experience (Lyness & Thompson, 1997; Morrison & Von Glinow, 1990; Stroh, Brett, & Reilly, 1992).

More recently, some researchers have directly accounted for performance differences as explanations for reward differences among men and women. In the context of sales employees working for a single firm, Joshi and colleagues (2006) found that objective

sales performance (i.e., sales targets achieved) partially explained sex differences in annual salary after accounting for the sex composition of the work unit. In a more recent longitudinal analysis of personnel practices in an organization in the service industry, Castilla (2012) reported that sex and race differences in salary increases persisted after controlling for performance evaluations received by employees (see also Elvira & Town, 2001). In fact, researchers have found that reward systems that are tied to performance evaluations tend to exacerbate (rather than explain) inequity based on demographic attributes (Castilla, 2012; Castilla & Benard, 2010). Contrary to these findings, however, across employment settings, Blau and Devaro (2007) found that, after controlling for performance differences, there were no significant difference in wage growth between men and women. In order to empirically examine whether these performance differences explain gender inequality in organizations across multiple job contexts and to resolve discrepancies in prior findings, we posit that:

Hypothesis 5. Differences in organizational rewards can be fully accounted for by differences in performance evaluations between men and women.

Bundles of Contextual Factors Associated with Sex Differences in Performance Evaluations and Rewards

So far, based on extant research, we have proposed the independent effect of occupation-, industry-, and job-level variables on performance and reward differences between men and women. Based on past research, it is not clear whether two or more of these contextual attributes might operate together to explain sex differences in work-related outcomes. In order to develop a unified framework, it may also be worthwhile to consider whether these various attributes jointly influence sex differences in employment outcomes. For example, do gender-balanced occupations *and* industries with a high proportion of female executive jointly enable women to close the gender gap in organizational rewards? Are men more likely to outperform women while performing complex jobs in prestigious occupational contexts? In order to examine whether bundles of occupation-, industry-, and job-level contexts combine or complement each other to drive sex differences in performance and rewards, we also conducted an exploratory qualitative comparative analysis (QCA), as described below.

METHODS

Literature Search

We used computerized database searches of *PsycINFO*, *EBSCO*, and *ProQuest* to generate a pool of potential articles. To identify all of the articles that investigated gender and our outcomes of interest, we used combinations of the following search terms: “gender” (“sex,” “male/female”), and “task,” “job,” “in-role performance,” or reward outcomes such as “pay,” “promotion,” “salary,” “wages,” “rewards,” or “bonus.” We also manually searched major management journals in the field (Gomez-Mejia & Balkin, 1992), as well as disciplinary journals that focus on organizational research, to supplement the electronic search (e.g., *American Journal of Sociology*, *Industrial and Labor Relations Review*). To find unpublished sources such as dissertations and convention presentations, we searched Dissertation Abstracts International, and the programs of the annual meetings of the Academy of Management and the Society for Industrial and Organizational Psychology. We also contacted researchers in related areas to obtain current and unpublished studies that might fit our criteria for inclusion.

We used several inclusion criteria to select studies in the meta-analysis. First, we included studies that reported an *individual-level* effect of gender on performance evaluations or rewards. To calculate the standardized mean difference (d) between male and female groups, a study had to report the mean, standard deviation, and sample size for each gender group. If no such information was available, an appropriate statistic such as zero-order correlation, t , or F had to be provided to allow the computation of the standardized mean difference using formulas detailed by Lipsey and Wilson (2001). We also included studies that considered gender as a control variable as long as they reported relevant information with respect to performance and rewards (e.g., correlations) and transformed into the standardized mean difference. Because our study examined the moderating effects of occupation-, industry-, and job-level context on gender inequality, we focused primarily on studies that were conducted in field settings in which the effects of these variables were likely to manifest in sex differences in performance and rewards. We did not include in our dataset studies that relied on student samples, used experimental designs, or involved tasks in artificial environments (e.g., simulations).

Using all of the processes described above, the final sample set of our analyses yielded a total of 190 effect sizes and 474,732 individuals from 142 studies (73 for performance evaluations and 69 for organizational

rewards) conducted between 1985 and 2013.¹ This time period covers almost 30 years of research on the topic of gender in the management domain, and also represents the era following the civil rights legislation and the feminist movement, during which time women have made substantial gains in education, but have continued to face challenges with respect to gender parity in the workplace (Blau & Kahn, 2007; Reskin, 2003). From a practical standpoint, the reporting of correlations and other descriptive statistics was sparse in the research prior to the 1980s.

Coding and Variables

We thoroughly examined and coded all studies included in the sample based on a coding protocol (Lipsey & Wilson, 2001). We collected relevant information regarding effect size (d), key antecedents (occupation, industry, job level), and control variables from sample studies. We also collected data from secondary sources for our moderating variables (see below for a detailed explanation). Given that we used objective data for all of the variables included in the analyses, subjectivity or intercoder unreliability was not a concern; however, we also cross-checked our coding with all of the coders several times throughout the coding process.

Performance evaluations. Measures of *performance evaluations* indicated a rater's (mostly a supervisor's) evaluation of an employee's overall effectiveness and performance in fulfilling his or her everyday roles and responsibilities (e.g., Wayne & Liden, 1995; Williams & Anderson, 1991). We focused on job, task, or in-role performance (Borman & Motowidlo, 1997), and excluded studies that measured other dimensions such as contextual performance, citizenship behavior, or creative performance. When multiple measures of performance evaluations were available from a single study, we calculated a composite effect size by averaging all effect size information. We also coded the reliabilities of the measurement instruments whenever available. When the reliabilities were not reported, we used the average reliability of the same variable from all of the other studies in our dataset. The average reliability of the performance evaluation measure was .89.

Organizational rewards. Consistent with the conceptualization of tangible or actual rewards in the previous meta-analyses on this topic (Ng, Eby, Sorensen, & Feldman, 2005; Podsakoff, Whiting,

¹ A brief summary of studies included in the analysis is available from the first author on request.

Podsakoff, & Blume, 2009), in our study we defined *organizational rewards* as extrinsic forms of reinforcement offered by an organization (Byron & Khazanchi, 2012). Specific measures included salary level and salary increases, bonuses and incentive payments, and number of promotions. All of the studies in our dataset provided objective organizational rewards measures.

Proportion of men in an occupation. To obtain data regarding the *proportion of men in an occupation*, we referred to the U.S. Bureau of Labor Statistics (BLS) Current Population Survey. We first coded relevant information (occupations involved in the samples) based on sample descriptions of primary studies; we then obtained the occupational sex composition data from the BLS dataset and assigned this information to the occupations included in the studies as close to the year in which the primary study was conducted. For studies that included multiple occupations, we found sex composition data for each occupation category from the BLS and then calculated a composite value by averaging information on all participating occupations. However, we excluded from the analysis any studies that involved multiple occupations with varied sex composition (e.g., accounting and engineering in the same sample), involved non-U.S. samples, or were unclear about the samples ($n = 38$). Same coding rules were applied to coding other key variables.² Among the studies

included in our analysis, the percentage of males in various occupations ranged from 12% (i.e., bank tellers) to 88% (i.e., automotive dealers) and 97% (i.e., firefighters), with an average of 61%.

Occupational prestige. *Occupational prestige* refers to the consensual rating of an occupational category with respect to its worthiness, based on several socioeconomic factors such as income and the education levels of incumbents (Hauser & Warren, 1997). To code and obtain information for occupational prestige, we relied on the Nakao-Treas Socioeconomic Index (Nakao & Treas, 1994), which is based on a nationwide survey undertaken by the National Opinion Research Center (NORC). It provides prestige scores of a wide range of more than 500 occupation titles ranked on a 100-point scale. Specifically, it uses socioeconomic indices such as education and income levels that are standardized for full-time incumbents' age distributions. The prestige scores are then calculated based on the age-weighted average of education and income levels for each occupation. We first coded occupation titles based on the sample descriptions in primary studies and then assigned the corresponding occupational prestige score from the index. The mean of the Nakao-Treas prestige scores in our sample was 53.2—equivalent to a middle-class occupational prestige level. Examples of less prestigious occupations in our sample were seasonal sales clerks and warehouse workers, with scores of 27.7 and 22.1, respectively. Examples of the most prestigious occupations in our sample were executives and primary care physicians, with scores of 64.2 and 71.79, respectively.

Proportion of women at executive or senior manager levels within industry. To code the *proportion of women at executive or senior manager levels within industry*, an industry-level context variable, we consulted the upper level (executives and senior managers) female representation data for each industry available in the U. S. Equal Employment Opportunity Commission (EEOC) survey, *Job Patterns for Minorities and Women in Private Industry Survey*. Drawing on detailed information on the EEOC survey, which was based on the North American Industry Classification System (NAICS) two-digit industry code, we obtained data about the female representation in upper level positions for the industries identified and included in the sample. Once again, we obtained data as close to the timing of the primary studies as possible and used only U.S.-based samples. Examples of industries with fewer high-ranked women were utility (13%)

² In order not to oversimplify the coding of job, industry, and occupational variables, we took the following steps during the coding process. For primary studies that included samples within a single work context—e.g., primary care physicians (Hekman, Aquino, Owens, Mitchell, Schilpzand, & Leavitt, 2010), public school teachers (Ferris, Bergin, & Wayne, 1988), executive managers (Graves, Ruderman, Ohlott, & Weber, 2012), bank tellers (Schaubroeck & Lam, 2002), automotive dealers (Liu, Loi, & Lam, 2011), etc.—we were able to code the key variables directly. These studies constituted more than 50% of our data. Primary studies that relied on multiple organizations or work contexts often included occupations and jobs that were very similar, e.g., scientists and engineers (Keller, 2012), system analysts and programmers (Igbaria & Baroudi, 1995), etc. In these cases, we used average occupational gender composition, occupational prestige, or job complexity scores. However, we excluded from our regression analysis studies that provided relatively limited information about the sample or which involved multiple diverse work contexts—e.g., a mixed sample consisting of employees working in areas of finance, sewage, libraries, and childcare (Jepsen & Rodwell, 2009)—or non-U.S. samples.

and manufacturing (18%); on the other hand, the healthcare industry employed about 54% of upper level female managers, which was the highest female representation at executive levels in the sample.

Job complexity. We obtained *job complexity* scores using scores available at the *Occupational Network (O*Net)* database—a source that contains information on standardized and job-specific descriptions of the work activities, content, characteristics, and skills required for 975 job titles. Based on operationalizations in past research (Auster, 1989; Campbell, 1988), job complexity was assessed with the *O*Net* skill variable labeled “complex problem solving,” which indicates the abilities needed to solve novel, ill-defined problems in complex, real-world settings, and is typically higher for jobs that are characterized by greater discretion and autonomy. The *O*Net* values for complex problem-solving skill ranged from 0 to 100, with higher scores indicating the skill’s greater importance to the job. Our sample includes relatively less challenging and more routine jobs such as clerical workers (38) and bank tellers (47), as well as highly complex jobs such as senior scientists or engineers (85) and top executives (88).

Control variables. We also coded several control variables (study and sample characteristics) to enhance the robustness of the analyses. First, we used a dummy variable capturing whether the effect sizes were obtained from studies with longitudinal (coded as “1”) or cross-sectional (coded as “0”) designs. Publication status was also dummy coded (“1” if published). To take into account potential temporal effects, we coded the publication year of each primary study from which effect sizes were derived (Carney, Gedajlovic, Heugens, Van Essen, & Van Oosterhout, 2011). We coded average sample tenure from each study (the average number of years that individuals spent in the organization) because tenure can serve as a proxy for the human capital of employees included in the sample (e.g., Dipboye & Colella, 2005). We also considered rating source (i.e., supervisory versus peer ratings) as an additional control; however, it was not included in a final model because of a lack of studies using peer ratings in our data.

Meta-Analytic Procedures

We employed several types of analytical approach to the meta-analysis. First, to examine sex differences in performance and rewards, we calculated effect

size d , the standardized mean difference between female and male groups for continuous measures of performance evaluations and organizational rewards, using procedures suggested by Hedges and Olkin (1985). Measurement unreliability was corrected for subjective performance ratings (Lipsey & Wilson, 2001). We calculated confidence intervals of 95% to judge the statistical significance of the effect sizes (Whitener, 1990). To test the homogeneous distribution of the effect sizes (detecting possibilities of potential moderators), we also calculated the Q statistics (Hedges & Olkin, 1985).

Testing for potential publication bias. To address any potential publication bias issues, we generated contour-enhanced funnel plots (Kepes, Banks, McDaniel, & Whetzel, 2012). These graphs are scatterplots in which the effect sizes estimated from primary studies are displayed on the X axis and the inverse of a sample’s standard error (i.e., precision) along the Y axis (Kepes et al., 2012). Since the precision depicted on the Y axis increases as sample size increases, in the absence of publication bias, studies with larger (i.e., more precise) sample size are scattered toward the top, whereas studies with small sample size (thus less precise) are scattered widely toward the bottom of the funnel plot. If sampling error is not the only reason for variance in the sample distribution and studies with insignificant results are omitted (i.e., the presence of publication bias), asymmetrical funnel plots may result (Kepes et al., 2012); if the funnel plot is symmetrical, we may conclude that there is no publication bias.

Meta-regression analysis. To test our main hypotheses (Hypotheses 1–4), we conducted random effects meta-analytic regression analysis, which comprises a series of weighted least squares (WLS) regressions investigating the relationship between key independent variables and the effect estimate as the outcome variable (Lipsey & Wilson, 2001). In random effects meta-regressions, effect sizes are weighted by the differences in precision (inverse variance weights) (Hedges & Olkin, 1985). This approach also allows us to take into account residual heterogeneity not modeled by the independent variables, but arising out of unmeasured between-study differences (Lipsey & Wilson, 2001). We tested the effects of each of the occupation-, industry-, and job-level variables with control variables in separate models. Although meta-analytic regressions allow us to test the effect of multiple factors (i.e., all independent variables) at the same time in a full regression equation, this approach is

often not feasible because of insufficient studies across all variables (Thompson & Higgins, 2002).

Monte Carlo simulations for testing robustness of estimates. To check the robustness of our regression findings, we further conducted a permutation test based on Monte Carlo simulations (Higgins & Thompson, 2004). In meta-regressions, owing to relatively small sample sizes (especially when multiple covariates are considered), there is an increased chance of at least one false positive finding (i.e., of “Type I” error) (Harbord & Higgins, 2008). In permutation tests, the covariates are randomly allocated to the outcomes over multiple runs (1,000 runs in our analyses) and a *t*-statistic is calculated for each run. The true *p* value for the relationship between a given covariate and the outcome is computed by counting the number of times that these *t*-statistics are greater than or equal to the observed *t*-statistic. Thus the adjusted *p* value from a permutation test can be interpreted as the degree to which one may be surprised about the observed effect of a variable if all other covariates were also included in the meta-regression (Higgins & Thompson, 2004). A permutation test approach is highly recommended to assess statistical significance in meta-regressions (see Higgins & Thompson, 2004 for details).

Mediation test. To test whether sex differences in rewards can be explained by performance evaluation differences (Hypothesis 5), we conducted a semipartial correlation test based on the meta-analytic estimates calculated about $X \rightarrow Z$, $X \rightarrow Y$, and $Z \rightarrow Y$ (where *X* is gender, *Z* is performance evaluations, and *Y* is organizational rewards). (For a similar approach, see Gajendran & Harrison, 2007.) We first transformed our standardized mean difference (*d*) to correlation (*r*) using the equation provided by Lipsey and Wilson (2001). After controlling for the effect of performance evaluations, we calculated a semipartial correlation between gender and rewards using the harmonic mean as the sample size for the meta-analytic correlation (Viswesvaran & Ones, 1995). We used semipartial correlations instead of partial correlations because the former provides a correlation such that performance (*Z*) is partialled out of gender (*X*) alone, whereas a partial correlation coefficient has performance (*Z*) partialled out of both rewards (*Y*) and gender (*X*) (Lewis-Beck, Bryman, & Liao, 2004). Given that we were interested in the change in $X \rightarrow Y$ after partialling out $X \rightarrow Z$, not $Z \rightarrow Y$, the semipartial correlation could give us a precise statistic for testing mediation by not holding performance constant for rewards.

Exploratory qualitative comparative analysis to identify bundles of contextual factors. We applied a crisp set version of the QCA (Ragin, 2000; 2008) that has received growing attention in management research and which is ideal for identifying how multiple independent variables can jointly influence a phenomenon of interest (see Fiss, 2011; Greckhamer, Misangyi, Elms, & Lacey, 2008). Unlike other statistical procedures that rely on significance testing or aim at explaining variance in an outcome of interest, QCA identifies which specific set of causal attributes is common across all cases of an outcome. For example, the analysis could identify the combinations of occupational, industry, or job attributes that occur across cases of higher levels of sex differences in rewards or performance (Greckhamer et al., 2008). In our study, a “case” is an effect size (*d*) derived from primary studies. We applied QCA in this study as an exploratory approach for identifying whether multiple attributes (i.e., occupation-, industry-, and job-level variables) are associated with cases of high levels of sex differences in rewards or performance in the data. Qualitative comparative analysis relies on the principles of Boolean algebra, and utilizes binary data and combinatorial logic to identify the causally relevant conditions that explain a focal phenomenon (Ragin, 2008).

A crisp set QCA analysis is inherently dichotomous and evaluates each case in the data based on either membership or nonmembership of a set (see Greckhamer et al., 2008, for details). The cutoffs for dichotomizing attributes were based on theoretical considerations (Ragin, 2000). Conceptually, a *d* greater than zero implies a set of cases in which men receive higher evaluations or are rewarded more than women, while a *d* lower than zero represents cases in which women receive higher evaluations or are rewarded more than men. Although calibrating outcomes based on theory alone is ideal, practical considerations, such as the distribution of data, can also guide these decisions (Greckhamer et al., 2008). For performance differences, our data included a full range of effect sizes, with *ds* ranging from positive, near zero, to negative, with a median close to zero (i.e., *ds* ranged from -2.16 to 2.27 , with a median of $-.01$). Based on conventions followed in past research, as well as theoretical considerations, we defined a set of cases in which *ds* were positive as cases in which men received higher evaluations than women and other cases as nonmembers of this set. We also defined a set of cases in which *ds* were lower than the median value as

a set of cases in which women received higher evaluations than men and other cases as nonmembers of this set. For reward differences, d s were positively skewed, ranging from $-.37$ to 3.07 , with a median of $.65$. We used the median as a cut-off point to represent a set of cases in which men were rewarded higher than women and other cases as nonmembers of this set.

Each occupational, industry, and job attribute was also similarly calibrated, with cases above the median value coded as "1," meaning the presence of a specific condition, or "0," meaning nonmembers of this group. Based on median values, each of the contextual attributes was defined as follows: Occupations with more than 40% women were categorized as "balanced occupations," occupations with prestige scores of more than 60 were classified as "prestigious occupations," industries with more than 30% female executives or senior managers were categorized as "high proportion of women in executive/senior manager levels," and jobs with complexity scores over 70 were classified as "complex jobs."

Once the data were coded into crisp sets, the second phase of the QCA approach involved examining the sufficiency of the four attributes (occupational demographic composition, occupational prestige, percentage of female executives in the industry, and job complexity) for explaining the four outcomes: Men evaluated higher than women; women evaluated same as, or higher, than men; men rewarded higher than women; women rewarded same as, or higher, than men. Based on the truth table algorithm provided by the QCA software, configurations for each of the outcomes were specified. In the current analyses, configurations that yielded at least three cases with consistency of more than 1.00 were included in the final analysis. These cut-off values are also based on conventions followed in past QCA research. Our decision to include only configurations with at least three cases allowed us to include 80% of the cases in the analyses (see Misangyi & Acharya, 2014; Ragin & Rihoux, 2009). We used the most stringent cut-off of 1.00 for the consistency of the cases represented by that configuration (Ragin, 2000). Consistency is a measure of sufficiency of a particular configuration of attribute, and it indicates how closely a configuration is associated with an outcome. A consistency of 1.00 indicates that every time a particular combination of attributes is present, the outcome is also present. In other words, a configuration was included in the final results only when

100% of cases displaying a specific configuration of occupation-, industry-, or job-level attributes also displayed the outcome of interest (Ragin, 2008).

RESULTS

Sex Differences in Performance Evaluations and Organizational Rewards

Using the meta-analytic techniques described above, we first examined the magnitude of sex differences in performance evaluations and rewards (see Table 1). For performance evaluations, the mean effect size corrected for unreliability was positive (i.e., men outperformed women), but the confidence interval included zero ($d = .04$, $k = 93$, 95% confidence interval, or CI = $-.12-.20$), indicating that there were no significant sex differences in performance evaluations. For organizational rewards, results indicated that men received significantly higher pay and promotions than women ($d = .56$, $k = 97$, 95% CI = $.45-.65$).³ The mean sex difference in rewards was 14 times larger than the mean sex difference in performance evaluations. Table 1 also shows that there is considerable heterogeneity among the effect sizes, as indicated by the Q statistic. The Q values for both performance evaluation and reward outcome measures were highly significant ($p < .01$), indicating that the effect sizes vary across the studies and that potential moderators play a role.

To check potential publication bias in our results, we also plotted contour-enhanced funnel plots for both performance and reward outcomes. In the plot, the white area is where statistically insignificant effect sizes are located. The darkest and thinnest area is where marginally significant effect sizes lie ($.05 < p < .10$). The thin and dark gray shaded, and the large light gray shaded, areas are where statistically significant effect sizes are found (Kepes et al., 2012). For performance evaluations, the plot is symmetrical (see Figure 1a); also the dots, which

³ Since we grouped together various types of reward (i.e., pay, bonus, and promotion) into a broad category of organizational rewards, we also undertook additional subgroup analyses to see whether there were any differences based on the type of reward. The overall patterns were consistent, and we found that sex differences were larger for pay ($d = .67$, $k = 67$, 95% CI = $.55-.75$) than for promotions ($d = .24$, $k = 29$, 95% CI = $.08-.39$). Owing to the lack of samples, we were not able to conduct the separate analyses for each subgroup using more specified types of reward.

TABLE 1
Sex Differences in Performance Evaluations and Organizational Rewards

Outcomes	<i>k</i>	<i>n</i>	Effect size <i>d</i>	95% CI	<i>Q</i>
Performance evaluations	93	95,882	0.04	−0.12, 0.20	53,545.10*
Organizational rewards	97	378,850	0.56	0.45, 0.65	207,187.50*

Note: *k* = no. of effect sizes; *n* = total no. of individuals counted by effect sizes; effect size *d* = sample size weighted mean effect size corrected for unreliability; 95% CI = 95% of confidence interval of *d*; *Q* = effect size heterogeneity statistic indicating the possibility of moderators

**p* < .01

represent studies included in the analysis, are evenly distributed across the shadowed areas (i.e., regions of statistical significance), as well as the white area (i.e., the area of nonsignificance). For rewards (see Figure 1b), the plot is less symmetrical, indicating the possibility of the skewedness of our samples. However, a closer look at the plot reveals that the dots are allocated across regions of statistical significance, as well as regions of nonsignificance (the white area). Moreover, potentially “missing” studies are more likely to be located on the bottom left-hand side of the plot, which is the region representing statistical significance, but lower sample sizes or higher standard errors. (Note that there are almost no dots in that area.) Thus, for both performance and reward outcomes, we did not find strong publication bias issues in the sample.

Results of Meta-Analytic Regressions

Table 2 summarizes results from a series of meta-analytic regressions for performance evaluations (Models 1–3) and reward differences (Models 4–6).⁴ Overall, we found general support for the hypotheses. Hypotheses 1a and 1b predicted that sex differences in performance evaluations and rewards would increase as the proportion of men in an occupational

category increased. In support for the hypotheses, we found significant positive relationships between the proportion of men in an occupation and performance differences (Model 1: $b = 1.37$, $p < .01$), as well as reward differences (Model 4: $b = 2.08$, $p < .01$), between men and women.

Our second set of hypotheses (Hypotheses 2a and 2b) predicted that sex differences in performance evaluations and rewards would increase as the prestige level of the occupational context increases. Models 1 and 4 also present the regression results testing these hypotheses. Contrary to our expectations, we did not find a significant relationship between occupational prestige and sex differences in performance evaluations (Model 2: $b = .03$, $p > .10$); however, sex differences in organizational rewards increased with occupational prestige (Model 4: $b = .06$, $p < .05$), rendering support for Hypothesis 2b. Both regression models were statistically significant (Model 1: $F = 5.68$, $p < .05$; Model 4: $F = 6.15$, $p < .01$), and the adjusted R^2 also indicated that more than 20% of between-study variance can be explained by occupational sex composition and prestige, along with other control variables.

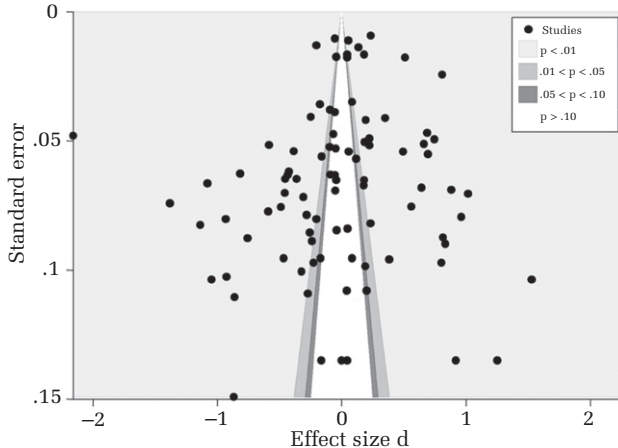
Hypotheses 3a and 3b predicted that sex differences in performance and rewards would be smaller in industries with a higher representation of women at the executive and senior managerial levels. Models 2 and 5 in Table 2 represent these findings. Providing strong support for our hypotheses, results revealed that there were significant negative relationships between the representation of women at higher levels within industry and performance evaluation differences (Model 2: $b = -2.80$, $p < .01$), as well as reward differences (Model 5: $b = -3.34$, $p < .01$). Both regression models were significant (Model 2: $p < .01$, adjusted $R^2 = .18$; Model 5: $p < .01$, adjusted $R^2 = .23$).

As posited in Hypotheses 4a and 4b, we found that job complexity had a positive effect on sex differences in performance evaluations and rewards. Findings indicated that men received higher performance evaluations as job complexity increased (Model

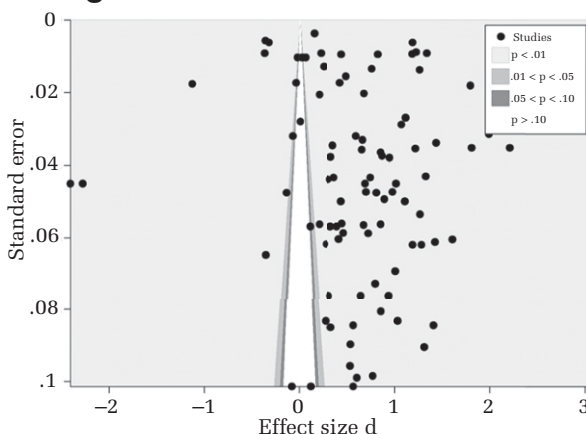
⁴ We present results for occupation-, industry-, and job-level variables, along with control variables in this section. As a result of the listwise deletion of missing cases, including all variables in a single meta-regression was not feasible. However, we undertook additional analyses that included various combinations of occupation- and industry-level variables simultaneously to examine whether the overall pattern of results remained the same and explained additional overall variance in the effect sizes. For both performance evaluation and reward outcomes, we found consistent patterns of results in our multiple regressions. Variance inflation factors for these analyses were well below the cut-off value of 10. We also present results of Monte Carlo simulations that address any concerns of Type I errors.

FIGURE 1
Funnel Plot for Studies Testing Sex Differences in
Performance Evaluations and Organizational
Rewards

A. Performance Evaluations



B. Organizational Rewards



3: $b = .06$, $p < .01$). The relationship between job complexity and reward differences between men and women also revealed a similar pattern (Model 6: $b = .05$, $p < .01$). Both regression models were also statistically significant (Model 3: $p < .01$, adjusted $R^2 = .17$; Model 6: $p < .01$, adjusted $R^2 = .23$).

Results of Monte Carlo simulations. Based on Monte Carlo simulations, we conducted permutation tests to ascertain the robustness of these results. After 1,000 random runs based on Monte Carlo simulations, adjusted p values for significant effects remained significant ($p < .05$ for industry upper level female composition with performance evaluation differences; $p < .01$ for all others) after accounting for multiplicity (multiple testing), indicating the robustness or low probability of Type I error in our findings. For example, the adjusted

p value of .008 for occupational prestige with reward differences (more conservative than $p = .003$, as per Model 6) indicates a lower than 1% probability of false positive error; the adjusted p value for industry upper-level female composition with performance differences increases from .006 to .024 after 1,000 random runs, but it still remains significant ($p < .05$).

Mediation Test: Semipartial Correlation

To test whether sex differences in performance evaluations explain sex differences in organizational rewards (Hypothesis 5), we conducted a semipartial correlation test. Table 3 reports a semipartial correlation between sex and organizational rewards after controlling for sex differences in performance evaluations. Results indicated that performance evaluations did not appear to mediate the effects of sex on organizational rewards when comparing the semipartial correlation ($r = .27$, $p < .01$) with the original correlation ($r = .27$, $p < .01$). Sex differences in rewards remained significant after partialling out the effects of performance evaluations.

Results of Exploratory QCA Analyses

Table 4 presents the findings of a crisp set QCA analysis.⁵ The configurations are based on conventions developed in recent research (see Fiss, 2011; Greckhamer et al., 2008). Central contextual attributes are represented by • if present and ⊗ if absent. Peripheral attributes are represented by • if present and ⊗ if absent. Central conditions do not require

⁵ The following are illustrations of each of the four configurations from our data: Panel 1—Automotive dealers in the automobile retail industry engaged in direct customer sales (both the occupation and industry executive ranks are male-dominated and the occupation is rated as less prestigious); Panel 2:1—Bank tellers working in the retail banking industry (jobs are routine and the occupation is balanced, but relatively few women are represented at executive levels in the industry); Panel 2:2—Security employees in the retail services industry (e.g., malls), which task mostly entails assisting mall customers (the occupation is male-dominated, but there is a relatively high proportion of women at the executive levels in this industry); Panel 3—Engineering consultants working in the utility industry (the occupation is prestigious and male-dominated, senior executive ranks in the utility industry are also male-dominated, and tasks are dynamic and complex in nature).

TABLE 2
Results of Meta-Analytic Regressions

	Performance evaluations			Rewards		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Constant	−0.61 (0.36) [†]	0.82 (0.43) [†]	−1.29 (0.49) [*]	−0.69 (0.47)	1.45 (0.38) ^{**}	−2.50 (0.76) ^{**}
<i>Controls</i>						
Study design (1 = longitudinal)	−0.06 (0.24)	−0.14 (0.26)	−0.06 (0.25)	−0.04 (0.20)	−0.03 (0.18)	−0.07 (0.19)
Publication year	−0.02 (0.01)	−0.00 (0.01)	−0.01 (0.01)	−0.03 (0.01) [†]	−0.00 (0.01)	−0.01 (0.01)
Average sample tenure	0.02 (0.02)	0.03 (0.03)	0.02 (0.02)	−0.03 (0.03)	−0.01 (0.02)	−0.02 (0.03)
<i>Predictors</i>						
Proportion of men in occupation	1.37 (0.40) ^{**}			2.08(0.65) ^{**}		
Occupational prestige	0.03 (0.01)			0.06 (0.01) [*]		
Proportion of women in executive or senior manager levels within industry		−2.80 (0.74) ^{**}			−3.34 (0.85) ^{**}	
Job complexity			0.06 (0.01) ^{**}			0.05 (0.01) ^{**}
<i>k</i>	58	54	61	71	44	74
Adjusted <i>R</i> ²	0.21	0.18	0.17	0.22	0.23	0.23
Model <i>F</i>	5.68 [*]	3.79 ^{**}	3.90 ^{**}	6.15 ^{**}	4.13 ^{**}	6.49 ^{**}

Note: Unstandardized regression coefficients are presented; numbers in parentheses are standard errors; *k* = total no. of effect sizes involved in the analysis; the publication status (1 = published) dummy was dropped from the analysis owing to multicollinearity

[†] *p* < .10

^{*} *p* < .05

^{**} *p* < .01

counterfactual analyses and can be fully determined by the cases in the data. Peripheral conditions, on the other hand, require counterfactual analyses, and their inclusion in the solution is based on theoretically derived assumptions. Counterfactual analyses require all possible configurations (in our study, 2^k where *k* = four attributes) to be populated with cases, although it is often unlikely that all logically possible configurations can be populated in real organizational contexts (Ragin, 2008).

Further, each solution may consist of multiple configurations of attributes, and “coverage” of each configuration shows the proportion of consistent cases that display that particular configuration of attributes. Unique coverage is the proportion of cases that display only that particular configuration. In addition, the analysis also provides the overall consistency and coverage for all of the configurations in a particular solution. Table 4 shows three panels displaying solutions for each of the outcomes of interest. We excluded one outcome from the analyses—women rewarded the same, or higher, than men—because there were insufficient cases in the data. As shown in Panel 1, one configuration was sufficient for men receiving higher performance evaluations than women. Men received higher evaluations than women in male-dominated occupations (i.e., absence of gender-balanced

occupation) that were less prestigious (e.g., automotive sales, assembly line manufacturing) and had few women at the executive levels in the industry regardless of whether the job was complex or routine.

Panel 2 shows that two configurations were sufficient for women receiving higher performance evaluations than men. Across both configurations, note that “job complexity” is clearly absent. Further, gender balance in the occupation and higher proportion of women at senior executive or senior manager levels appear to be substitutes for one another: When one is present, the other is absent, and vice versa. Therefore it appears that women employed in routine jobs are likely to receive higher evaluations than men when either the occupation in which they are working is balanced or the industry as a whole has a higher representation of women at executive levels. These combinations apply regardless of occupational prestige.

Panel 3 shows that one configuration was sufficient for men receiving higher rewards than women. Men received higher rewards than women in prestigious occupations in which jobs were complex and women were underrepresented in senior positions at the industry level. Whether the occupation was balanced or not was irrelevant. Overall, across both performance evaluation and reward outcomes,

TABLE 3
Mediation Test: A Semipartial Correlation

Outcome	Estimated <i>r</i>	Semipartial correlation after controlling for differences in performance evaluations
Organizational rewards	0.27** (378,850)	0.27** (60,259)

Note: Estimated *r* indicates a correlation corrected for unreliability; significance of a semipartial correlation was evaluated against the harmonic mean of sample sizes (in parentheses)

** $p < .01$

it appears that higher representation of women at executive levels in the industry (presence or absence) was a common condition. Gender balance at the occupational level and job complexity emerged as common conditions in three out of the four configurations across all outcomes.

DISCUSSION

Nearly five decades after the passage of the Civil Rights Act of 1964, academics and policymakers are still asking the question: Can women close the gap in performance and rewards in the workplace? The answer to this question based on our research is: Women often close the gap in performance, but not the gap in rewards. Across occupations ranging from bank tellers to accountants, industries ranging from IT to healthcare, and jobs ranging from mundane to challenging, our results show that sex differences in organizational rewards were almost 14 times larger than sex differences in performance evaluations. Moreover, performance differences did not explain reward differences between men and women. The percentage of men in an occupation and the complexity of jobs performed enhanced the male–female gap in performance and rewards. In highly prestigious occupations, women performed equally, but were rewarded significantly lower than men. Only in industries with a higher proportion of female executives did women reverse the gap in performance evaluations *and* in rewards.

Theoretical Implications: Identifying Pathways to Gender (In)Equity in Organizations

Our meta-analysis provides broad insights into the extent to which reward and performance differences vary based on various macro-level attributes of the work context, and we suggest below how future studies might further unpack pathways through which these contextual factors shape the

allocation of rewards and performance evaluations among men and women.

A noteworthy macro–micro linkage in our findings is represented by the significant effects of fairly distal contextual attributes—occupational sex composition and prestige—on differences in performance evaluations and rewards among men and women. In male-dominated occupations, it is possible that the masculine cultural norms lay the groundwork for how performance is evaluated and rewards are allocated. For example, in occupations such as offshore drilling or firefighting, overall cultural norms are likely to favor men, and status is likely to be confounded with gender. In these settings, women’s skill attainment may not offset the overall rewards gap because gender remains a dominant status cue and other status signals, such as educational expertise, are often discounted. Clearly, in these situations, women face barriers to translating any human capital gains into greater rewards at work (e.g., Joshi, 2014). We surmise that one important pathway to gender inequality in these settings is the interactional dynamics with peers and supervisors that often drive performance evaluations and reward allocation decisions (see also Joshi & Knight, 2015). We note that an increasing number of organizations are adopting “high involvement” performance management systems that rely on peer evaluations, which are also tied to reward allocation (Cappelli, 1999). These practices place a premium on social interactions among employees, and the extent to which their peers and supervisors recognize and utilize their expertise can be particularly consequential for women’s career advancement opportunities in these contexts (Joshi, 2014; Joshi & Knight, 2015; Ridgeway, 1991). Indeed, scholars have noted that these “strategic,” or “high involvement,” performance management and compensation practices serve to enhance, rather than reduce, gender inequality in organizations (e.g., Cappelli, 1999; Castilla, 2012). We join these scholars in calling for more research on how these high-involvement HR practices serve as

TABLE 4
Results of Qualitative Comparative Analysis

	Panel 1 Men receive higher performance evaluations than women	Panel 2 Women receive higher performance evaluations than men	Panel 3 Men receive higher rewards than women
	1	1 2	1
<i>Occupational context</i>			
Balanced occupation	⊗	●	⊗
High prestige occupation	⊗		●
<i>Industry context</i>			
High proportion of women in executive or senior manager levels	⊗	⊗	●
<i>Job context</i>			
High job complexity		⊗	⊗
Consistency	1.00	1.00	1.00
Raw coverage	0.25	0.20	0.20
Unique coverage	0.25	0.20	0.20
Overall solution consistency	1.00	1.00	1.00
Overall solution coverage	0.25	0.40	0.14

Note: Central conditions are represented by ● (presence) and ⊗ (absence); peripheral conditions are represented by ● (presence) and ⊗ (absence); threshold for consistency was 1.00; frequency was three cases/configuration; no configurations were sufficient for observing women receiving equal or higher rewards than men

mechanisms driving gender inequality and on whether these practices have different outcomes in traditionally male-dominated versus gender-integrated settings.

A particularly striking pattern in our findings is that, in highly prestigious occupations, women did not perform at lower levels than men, and yet men were rewarded significantly higher than women. We propose that, in these settings, it is particularly important to disentangle reward allocation decisions from performance evaluation decisions. In prestigious settings, employment practices may serve as hierarchy-enhancing agents that maintain social hierarchies facilitating male dominance (Sidanius et al., 1996). Indeed, professions such as academia and law offer particularly interesting avenues for further inquiry: In these high-prestige settings, performance criteria tend to be objective (e.g., billable hours or research productivity), yet reward allocation decision making is highly subjective, opaque, and adversarial, and often involves high stakes (many such settings having “up or out” promotion norms). We surmise that this discrepancy in performance evaluations and reward allocation decision making in prestigious contexts is an important mechanism explaining inequity in rewards. Future research could shed light on whether the adoption of promotion practices that do not involve zero-sum choices (e.g., law firms that offer nonequity partnerships) attenuates the social dominance orientation of key decision makers, and related

gender-based attributions of skills and competence, during performance or promotion reviews.

Our findings regarding the effects of job complexity and the sex composition of the upper echelons at the industry level also provide additional avenues for future research. We found that job complexity enhances sex differences in rewards and performance because complex jobs often entail ambiguous performance criteria and idiosyncratic roles that are less open to scrutiny. We propose that future research further unpack how job complexity, via mechanisms such as social comparisons and scrutiny among incumbents, predicts sex differences in employment outcomes. Reinforcing our meta-regression results, our QCA analyses revealed that both configurations representing situations in which women received higher evaluations than men involved less complex jobs (e.g., bank tellers). These routine jobs allow women to select shifts that are conducive to maintaining a work–life balance, are less likely to involve variable time-related demands, and do not place value on putting in “face time” outside of work hours. Even when the occupation was male-dominated or the industry-level executive ranks were predominantly male, our data revealed that women received higher performance evaluations than men while performing routine jobs. These findings underscore the importance of job structure as a mechanism explaining gender inequality, and call for more research into how job design can mitigate sex

differences in performance evaluations, bonus allocations, and promotion rates (see also Correll et al., 2014).

Our findings regarding the effects of the proportion of female executives in specific industry categories further highlight the importance of scrutiny and monitoring for reducing gender inequality. We found that women were rewarded at higher levels than men *and* received higher performance evaluations in only one setting: industries with a high proportion of female executives. The industry environment—specifically, the proportion of female executives—reflects institutional pressures driving the diffusion of egalitarian practices that also support greater monitoring of wage-setting and performance-related practices within firms (Dobbin, 2009). Thus the potential for scrutiny within firms may be driven by these industry-level factors, with important implications for sex differences in promotion and turnover rates within firms. Apart from the potential for scrutiny and monitoring, the proportion of women in executive positions at the industry level also represents greater access to social capital among women. For women, this access to industry-level networks can also help to identify alternative job opportunities within the industry, thereby improving the potential to negotiate a favorable employment contract. While past research has primarily focused on the implications of women's exclusion from social networks within organizations (e.g., Ibarra, 1992), we propose that future studies also examine whether the availability of extraorganizational networks facilitates women's advancement within firms across industry settings.

Caveats and Limitations

Our meta-analysis provides a comprehensive overview of sex differences in employment outcomes during a 30-year period and presents unique insights into the effects of various contextual factors influencing gender equity within firms—but it also has several limitations. First, although the overall sample size was fairly large, the meta-regression analyses involved relatively small samples. Our sample was restricted because some studies in our sample did not provide specific information to code, and we excluded several studies that involved multiple, very diverse work settings from the meta-regression analyses. Only studies that had exact information to match external data sources (e.g., BLS datasets, *O*Net*) were included in our final analyses. Owing to

these restrictions, we could not test an overall model including all of our key and control variables at once.

Second, we were also limited in testing the influence of other potentially important contextual variables on gender-based inequity in work outcomes. For example, organization-level variables such as culture, business strategy, or HR policies have an important influence on sex differences in performance and rewards, because they can shape employees' gender-based attributions and the discretionary behaviors of managers (see Gelfand, Raver, Nishii, & Schneider, 2005, for a review). Moreover, new structuralist perspectives in sociology have examined specific organizational features such as the formalization of procedures (e.g., Elvira & Graham, 2002), implementation of merit-based pay plans (Castilla, 2012), and restructuring (Dencker, 2008) in relation to sex differences in rewards. These studies, unlike ours, focus on single organizational contexts and provide more granular insights into mechanisms governing gender inequality. We could not incorporate these variables in the present model because of the lack of consistent information across the primary studies. Certainly, one avenue for future research may be to examine the effects of these organizational attributes on sex differences in performance or rewards within a broader context, such as in a highly prestigious occupation or in a specific industry context.

Another important limitation of our study is that, given our meta-analytic approach, we were unable to include fine-grained human capital controls such as labor market experience or educational level of the sample. Although we find that sex differences in rewards persisted even after accounting for performance differences and we controlled for the overall tenure of the sample, this is a critical gap. We note that the duration of studies included in our meta-analysis involves a period in which women have more than closed the gap in education and skills attainment. We therefore hope that our findings provide additional explanations for continued gender inequality in the workplace.

On a related note, we were unable to isolate “selection effects” or ascertain what types of process led men and women to choose specific jobs or occupations. For example, research shows that women often opt out of certain jobs even though they perform them as well as men because they differ from many men in their risk-seeking, competitive, and social preferences (Croson & Gneezy, 2009; Niederle & Vesterlund, 2007). It is therefore likely that the women who do choose to stay in high-prestige occupations or complex jobs are markedly different

from others based on some unmeasured attribute(s). We were not able to account for these variables characterizing men and women in our sample. We propose that future meta-analyses compare effect sizes from studies that generate selection-biased estimates (i.e., based on ordinary least squares, or OLS, analysis) with those from selection-corrected models (e.g., models using instrumental variables) across various contexts to examine how these estimates may vary. Additionally, future research may extend the argument developed here to other types of work outcome, such as contextual performance or creativity, or withdrawal behaviors such as turnover, and also examine whether sex differences vary for specific types of reward outcome (e.g., salary increases versus bonuses or promotions). We could not include those variables in the current study because of data limitations, as well as theoretical parsimony.

Finally, while the examination of our funnel plots did not reveal serious publication bias issues, we acknowledge that—especially on a topic such as gender—sometimes not only do null results get filed away, but also politically incorrect results (e.g., “women consistently obtain more rewards than men”). This tendency may yield biased estimates in the sample and cause a potential “file drawer” problem for the meta-analysis.

Implications for Diversity Management

Despite the growing adoption of diversity management practices, research has found that many of these practices (such as mentoring, network groups, or diversity training) have no direct benefits for enhancing access for underrepresented demographic groups within firms and, in fact, that some diversity management practices may even be detrimental to the advancement of underrepresented groups (Kalev et al., 2006). Our findings offer some practical insights that might mitigate gender-based inequity across various work settings and help to tailor diversity management practices. Broadly, we propose that diversity management practices focus on three issues: integrating accountability structures into performance management and compensation practices, designing jobs to promote greater equity among incumbents, and implementing industry-wide mentoring programs for women.

It appears that the potential for scrutinizing performance management and reward allocation processes is one mechanism by which various contextual factors may influence sex differences

in rewards or performance. We propose that accountability structures be built into performance management and reward allocation processes to formalize the scrutiny of these practices, particularly in contexts that are more susceptible to gender inequality. Research shows that the presence of compliance or affirmative action officers has positive effects on the hiring and promotion of women and minorities in organizations (e.g., Edelman, 1992). These findings may be extended to other practices as well. Although it may not be possible to make evaluative or compensation-related decision making fully transparent to employees in many work settings, the presence of neutral third-party observers or advisors can introduce checks and balances to avoid any systematic differences based on gender. For example, in many organizations, managers hold “calibration meetings” to allocate performance ratings to all of their direct reports and then allocate bonuses according to these ratings. Trained neutral observers or advisors present at these meetings may be able to direct discussion away from decision making based on stereotypes or biases toward performance-related information.

A second implication of the effects of job context in our study is that organizations seeking to address inequity introduce interdependence among incumbents in job categories that are more complex or ambiguous in their structure. Although task interdependence among employees within jobs is common, outcome interdependence, such as bonuses based on collective performance, may be critical in these contexts. This type of interdependence could motivate coworkers to seek individuating information regarding underrepresented groups, thereby avoiding reliance on stereotype-based cues of expertise or competence (Fiske, 1998). Complex jobs can also be designed to factor in the time spent in after-hours socializing or face time with clients, so that employees with caring responsibilities are not disadvantaged. Consider, for example, the experience of a major professional accounting firm that found that changing the job requirement to be present at the client site from five days a week to three days a week led to greater engagement and lower turnover among women *and* men in the firm, without any detrimental effects on firm revenues (McCracken, 2000).

Finally, our finding regarding the effects of industry-wide female representation at executive levels is also noteworthy. Given that mentoring programs within organizations have been unsuccessful in bringing about women’s advancement (Ibarra, Carter, & Silva, 2010), we propose that industry-wide mentoring

groups supplement ongoing career development among women. These network groups can be a source of mentoring and advice from senior-level women, industry-wide, as well as directing women toward opportunities for further skill and personal leadership development.

CONCLUSION

Although gender inequality is widespread, a close examination of management research on sex differences in performance evaluations and rewards showed that these effects also vary considerably across work contexts. We proposed that these variations offer fruitful avenues for building an actionable theory of gender inequality in organizations. This meta-analysis provides some important heuristics to identify contexts in which sex differences in performance evaluations and rewards are more or less likely. In an era in which gender bias or discrimination is rarely overt, or even intentional, identifying the sources of chronic gender inequality offers a compelling, yet challenging, agenda for management research. We call for further theoretical developments in the management domain to identify how, in specific work contexts, seemingly neutral organizational practices shape the structure of work, the definition and evaluation of performance, and the allocation of rewards to reinforce gender inequality.

REFERENCES

References marked with an asterisk (*) indicate studies included in the meta-analysis.

- *Abraham, J. D., & Morrison, J. D., Jr. 2003. Validity study relationship between the performance perspectives inventory's implementing scale and job performance of incumbents in a dual security/service role. *Applied H.R.M. Research*, 8: 73–76.
- Acker, J. 2006. Inequality regimes: Gender, class, and race in organizations. *Gender & Society*, 20: 441–464.
- *Allen, T. D. 2001. Family supportive work environments: The role of organizational perceptions. *Journal of Vocational Behavior*, 58: 414–435.
- *Allen, T. D. 2006. Rewarding good citizens: The relationship between citizenship behavior, gender, and organizational rewards. *Journal of Applied Social Psychology*, 36: 120–143.
- *Ang, S., Van Dyne, L., & Begley, T. M. 2003. The employment relationships of foreign workers versus local employees: A field study of organizational justice, job satisfaction, performance, and OCB. *Journal of Organizational Behavior*, 24: 561–583.
- *Aquino, K., & Bommer, W. H. 2003. Preferential mistreatment: How victim status moderates the relationship between organizational citizenship behavior and workplace victimization. *Organization Science*, 14: 374–385.
- *Arvey, R. D., Miller, H. E., Gould, R., & Burch, P. 1987. Interview validity for selecting sales clerks. *Personnel Psychology*, 40: 1–12.
- Auster, E. R. 1989. Task characteristics as a bridge between macro- and micro-level research on salary inequality between men and women. *Academy of Management Review*, 14: 173–193.
- Baron, J. N., & Pfeffer, J. 1994. The social psychology of organizations and inequality. *Social Psychology Quarterly*, 57: 190–209.
- *Barsness, Z. I., Diekmann, K. A., & Seidel, M. D. L. 2005. Motivation and opportunity: The role of remote work, demographic dissimilarity, and social network centrality in impression management. *Academy of Management Journal*, 48: 401–419.
- Bartol, K. 1999. Gender influences on performance evaluations. In G. Powell (Ed.), *Handbook of gender and work*: 165–178. Thousand Oaks, CA: Sage.
- *Bergeron, D. M., Shipp, A. J., Rosen, B., & Furst, S. A. 2013. Organizational citizenship behavior and career outcomes: The cost of being a good citizen. *Journal of Management*, 39: 958–984.
- Bergmann, B. 1974. Occupational segregation, wages, and profits when employers discriminate by race or sex. *Eastern Economic Journal*, 1: 103–110.
- Bernhardt, A., Morris, M., & Handcock, M. 1995. Women's gains or men's losses? A closer look at the shrinking gender gap in earnings. *American Journal of Sociology*, 101: 302–328.
- *Bertrand, M., & Hallock, K. F. 2001. The gender gap in top corporate jobs. *Industrial & Labor Relations Review*, 55: 3–21.
- *Bhatnagar, J. 2014. Mediator analysis in the management of innovation in Indian knowledge workers: The role of perceived supervisor support, psychological contract, reward and recognition and turnover intention. *International Journal of Human Resource Management*, 25: 1395–1416.
- *Bidwell, M. 2011. Paying more to get less: The effects of external hiring versus internal mobility. *Administrative Science Quarterly*, 56: 369–407.
- *Bizzi, L. 2013. The dark side of structural holes: A multilevel investigation. *Journal of Management*, Advance online publication. [DOI: 10.1177/0149206312471385].
- Blau, F. D., & Devaro, J. 2007. New evidence on gender differences in promotion rates: An empirical analysis of a sample of new hires. *Industrial Relations*, 46: 511–550.

- Blau, F. D., & Kahn, L. M. 1981. Race and sex differences in quits by young workers. *Industrial & Labor Relations Review*, 34: 563–577.
- Blau, F. D., & Kahn, L. M. 2007. The gender pay gap: Have women gone as far as they can? *The Academy of Management Perspectives*, 21: 7–23.
- *Bolino, M. C., & Turnley, W. H. 2003. Counternormative impression management, likeability, and performance ratings: The use of intimidation in an organizational setting. *Journal of Organizational Behavior*, 24: 237–250.
- Borman, W. C., & Motowidlo, S. J. 1997. Task performance and contextual performance: The meaning for personnel selection research. *Human Performance*, 10: 99–118.
- Bose, C., & Rossi, P. 1983. Gender and jobs: Prestige standings of occupations as affected by gender. *American Sociological Review*, 48: 316–330.
- *Brass, D. J. 1985. Men's and women's networks: A study of interaction patterns and influence in an organization. *Academy of Management Journal*, 28: 327–343.
- Briscoe, F., & von Nordenflycht, A. 2014. Which path to power? Workplace networks and the relative effectiveness of inheritance and rainmaking strategies for professional partners. *Journal of Professions and Organization*, 1: 33–48.
- *Byrne, Z. S. 2005. Fairness reduces the negative effects of organizational politics on turnover intentions, citizenship behavior and job performance. *Journal of Business and Psychology*, 20: 175–200.
- *Byron, K. 2007. Male and female managers' ability to read emotions: Relationships with supervisor's performance ratings and subordinates' satisfaction ratings. *Journal of Occupational and Organizational Psychology*, 80: 713–733.
- Byron, K., & Khazanchi, S. 2012. Rewards and creative performance: A meta-analytic test of theoretically derived hypotheses. *Psychological Bulletin*, 138: 809–830.
- Campbell, D. J. 1988. Task complexity: A review and analysis. *Academy of Management Review*, 13: 40–52.
- Cappelli, P. 1999. *The new deal at work*. Boston, MA: Harvard Business School Press.
- *Carless, S. A., & Arnup, J. L. 2011. A longitudinal study of the determinants and outcomes of career change. *Journal of Vocational Behavior*, 78: 80–91.
- *Carlson, D. S., Ferguson, M., Kacmar, K. M., Grzywacz, J. G., & Whitten, D. 2011. Pay it forward: The positive crossover effects of supervisor work–family enrichment. *Journal of Management*, 37: 770–789.
- *Carmeli, A., & Josman, Z. E. 2006. The relationship among emotional intelligence, task performance, and organizational citizenship behaviors. *Human Performance*, 19: 403–419.
- Carney, M., Gedajlovic, E. R., Heugens, P. P., Van Essen, M., & Van Oosterhout, J. H. 2011. Business group affiliation, performance, context, and strategy: A meta-analysis. *Academy of Management Journal*, 54: 437–460.
- Castilla, E. J. 2008. Gender, race, and meritocracy in organizational careers. *American Journal of Sociology*, 113: 1479–1526.
- *Castilla, E. J. 2012. Gender, race, and the new (merit-based) employment relationship. *Industrial Relations*, 51: 528–562.
- Castilla, E. J., & Benard, S. 2010. The paradox of meritocracy in organizations. *Administrative Science Quarterly*, 55: 543–576.
- Catalyst. 2008. Catalyst 2008 census of the Fortune 500 reveals women gained little ground advancing to business leadership positions. Accessed online at <http://www.catalyst.org/press-release/141/catalyst-2008-census-of-the-fortune-500-reveals-women-gained-little-ground-advancing-to-business-leadership-positions>.
- *Chatman, J. A., Polzer, J. T., Barsade, S. G., & Neale, M. A. 1998. Being different yet feeling similar: The influence of demographic composition and organizational culture on work processes and outcomes. *Administrative Science Quarterly*, 43: 749–780.
- *Chauvin, K. W., & Ash, R. A. 1994. Gender earnings differentials in total pay, base pay, and contingent pay. *Industrial & Labor Relations Review*, 47: 634–649.
- *Chen, Z. X., & Aryee, S. 2007. Delegation and employee work outcomes: An examination of the cultural context of mediating processes in China. *Academy of Management Journal*, 50: 226–238.
- *Chen, Z. X., & Francesco, A. M. 2003. The relationship between the three components of commitment and employee performance in China. *Journal of Vocational Behavior*, 62: 490–510.
- *Chen, Z. X., Lam, W., & Zhong, J. A. 2007. Leader-member exchange and member performance: A new look at individual-level negative feedback-seeking behavior and team-level empowerment climate. *The Journal of Applied Psychology*, 92: 202–219.
- *Chen, Z. X., Tsui, A. S., & Farh, J. L. 2002. Loyalty to supervisor vs. organizational commitment: Relationships to employee performance in China. *Journal of Occupational and Organizational Psychology*, 75: 339–356.
- *Chou, L. F., Cheng, B. S., Huang, M. P., & Cheng, H. Y. 2006. Guanxi networks and members' effectiveness in Chinese work teams: Mediating effects of trust networks. *Asian Journal of Social Psychology*, 9: 79–95.

- Cleveland, J. N., Vescio, T. K., & Barnes-Farrell, J. L. 2005. Gender discrimination in organizations. In R. L. Dipboye & A. Colella (Eds.), *Discrimination at work: The psychological and organizational bases*: 149–176. Mahwah, NJ: Lawrence Erlbaum Associates.
- Cohen, L. E., & Broschak, J. P. 2013. Whose jobs are these? The impact of the proportion of female managers on the number of new management jobs filled by women versus men. *Administrative Science Quarterly*, 2: 1–33.
- Cohen, L. E., Broschak, J. P., & Haveman, H. A. 1998. And then there were more? The effect of organizational sex composition on the hiring and promotion of managers. *American Sociological Review*, 63: 711–727.
- *Colquitt, J. A., LePine, J. A., Zapata, C. P., & Wild, R. E. 2011. Trust in typical and high-reliability contexts: Building and reacting to trust among firefighters. *Academy of Management Journal*, 54: 999–1015.
- Correll, S., Kelly, E., O'Connor, L. T., & Williams, J. 2014. Redesigning, redefining work. *Work and Occupations*, 41: 3–17.
- *Cropanzano, R., Rupp, D. E., & Byrne, Z. S. 2003. The relationship of emotional exhaustion to work attitudes, job performance, and organizational citizenship behaviors. *The Journal of Applied Psychology*, 88: 160–169.
- Croson, R., & Gneezy, U. 2009. Gender differences in preferences. *Journal of Economic Literature*, 47: 448–474.
- *Day, R., & Allen, T. D. 2004. The relationship between career motivation and self-efficacy with protégé career success. *Journal of Vocational Behavior*, 64: 72–91.
- Dencker, J. C. 2008. Corporate restructuring and sex differences in managerial promotion. *American Sociological Review*, 73: 455–476.
- *Diefendorff, J. M., Brown, D. J., Kamin, A. M., & Lord, R. G. 2002. Examining the roles of job involvement and work centrality in predicting organizational citizenship behaviors and job performance. *Journal of Organizational Behavior*, 23: 93–108.
- Dipboye, R. L., & Colella, A. 2005. *Discrimination at work: The psychological and organizational bases*. Mahwah, NJ: Lawrence Erlbaum Associates.
- DiTomaso, N., Post, C., & Parks-Yancy, R. 2007a. Workforce diversity and inequality: Power, status, and numbers. *Annual Review of Sociology*, 33: 473–501.
- DiTomaso, N., Post, C., Smith, D. R., Farris, G. F., & Cordero, R. 2007b. Effects of structural position on allocation and evaluation decisions for scientists and engineers in industrial R&D. *Administrative Science Quarterly*, 52: 175–207.
- Dobbin, F. 2009. *Inventing equal opportunity*. Princeton, NJ: Princeton University Press.
- *Dreher, G. F., & Ash, R. A. 1990. A comparative study of mentoring among men and women in managerial, professional, and technical positions. *The Journal of Applied Psychology*, 75: 539–546.
- *Dreher, G. F., & Cox, T. H. 2000. Labor market mobility and cash compensation: The moderating effects of race and gender. *Academy of Management Journal*, 43: 890–900.
- Eagly, A. H., Makhijani, M. G., & Klonsky, B. G. 1992. Gender and the evaluation of leaders: A meta-analysis. *Psychological Bulletin*, 111: 3–22.
- Eagly, A. H., & Karau, S. J. 1991. Gender and the emergence of leaders: A metaanalysis. *Journal of Personality and Social Psychology*, 60: 685–710.
- *Eby, L. T., Allen, T. D., & Douthitt, S. S. 1999. The role of nonperformance factors on job-related relocation opportunities: A field study and laboratory experiment. *Organizational Behavior and Human Decision Processes*, 79: 29–55.
- Edelman, L. B. 1992. Legal ambiguity and symbolic structures: Organizational mediation of civil rights law. *American Journal of Sociology*, 97: 1531–1576.
- Edelman, L. B., Fuller, S.F., & Mara-Drita, I. 2001. Diversity rhetoric and the managerialization of the law. *American Journal of Sociology*, 106: 1589–1641.
- EEOC. 2002. Job patterns for women and minorities in private industry. <http://www.eeoc.gov/stats/jobpat/jobpat.html>
- *Elfenbein, H. A., & O'Reilly, C. A. 2007. Fitting in: The effects of relational demography and person–culture fit on group process and performance. *Group & Organization Management*, 32: 109–142.
- *Elvira, M. M., & Cohen, L. E. 2001. Location matters: A cross-level analysis of the effects of organizational sex composition on turnover. *Academy of Management Journal*, 44: 591–605.
- *Elvira, M. M., & Graham, M. E. 2002. Not just a formality: Pay system formalization and sex-related earnings effects. *Organization Science*, 13: 601–617.
- Elvira, M., & Town, R. 2001. The effects of race and worker productivity on performance evaluations. *Industrial Relations: A Journal of Economy and Society*, 40: 571–590.
- Ely, R. J. 1994. The effects of organizational demographics and social identity on relationships among professional women. *Administrative Science Quarterly*, 39: 203–238.
- Ely, R. J. 1995. The power in demography: Women's social constructions of gender identity at work. *Academy of Management Journal*, 38: 589–634.

- Ely, R., & Padavic, I. 2007. A feminist analysis of organizational research on sex differences. *Academy of Management Review*, 32: 1121–1143.
- England, P. 1979. Women and occupational prestige: A case of vacuous sex equality. *Signs*, 5: 252–265.
- England, P., & McLaughlin, S. 1979. Sex segregation of jobs and male–female income differentials. In R. Alvarez & K. Lutterman (Eds.), *Associates: Discrimination in organizations*: 189–213. San Francisco, CA: Jossey-Bass.
- *Farh, C. I. C. C., Seo, M. G., & Tesluk, P. E. 2012. Emotional intelligence, teamwork effectiveness, and job performance: The moderating role of job context. *The Journal of Applied Psychology*, 97: 890–900.
- *Feldman, D. C., & Turnley, W. H. 2004. Contingent employment in academic careers: relative deprivation among adjunct faculty. *Journal of Vocational Behavior*, 64: 284–307.
- Felstiner, W., Abel, R., & Sarat, A. 1980. The emergence and transformation of disputes: Naming, blaming, claiming. *Law & Society Review*, 15: 631–654.
- *Ferris, G. R., Bergin, T. G., & Wayne, S. J. 1988. Personal characteristics, job performance, and absenteeism of public school teachers. *Journal of Applied Social Psychology*, 18: 552–563.
- *Ferris, G. R., Witt, L. A., & Hochwarter, W. A. 2001. Interaction of social skill and general mental ability on job performance and salary. *The Journal of Applied Psychology*, 86: 1075–1082.
- *Fields, D., Pang, M., & Chiu, C. 2000. Distributive and procedural justice as predictors of employee outcomes in Hong Kong. *Journal of Organizational Behavior*, 21: 547–562.
- Fiske, A. P. 1998. Human sociality. *International Society for the Study of Personal Relationships Bulletin*, 14: 4–9.
- Fiske, S. T., Bersoff, D. N., Borgida, E., Deaux, K., & Heilman, M. E. 1991. Social science research on trial: Use of sex stereotyping research in *Price Waterhouse v. Hopkins*. *The American Psychologist*, 46: 1049–1060.
- Fiss, P. C. 2011. Building better causal theories: A fuzzy set approach to typologies in organizational research. *Academy of Management Journal*, 54: 393–420.
- *Flynn, F. J., & Ames, D. R. 2006. What's good for the goose may not be as good for the gander: The benefits of self-monitoring for men and women in task groups and dyadic conflicts. *The Journal of Applied Psychology*, 91: 272–281.
- Gajendran, R. S., & Harrison, D. A. 2007. The good, the bad, and the unknown about telecommuting: Meta-analysis of psychological mediators and individual consequences. *The Journal of Applied Psychology*, 92: 1524.
- Gatta, M. L., & Roos, P. A. 2005. Rethinking occupational integration. *Sociological Forum*, 20: 369–402.
- Gelfand, M., Raver, J., Nishii, L., & Schneider, B. 2005. Discrimination in organizations: An organizational-level systems perspective. In R. Dipboye & A. Colella (Eds.), *Discrimination at work: The psychological and organizational bases*: 89–116. Mahwah, NJ: Lawrence Erlbaum Associates.
- Gerdes, E. P., & Garber, D. M. 1983. Sex bias in hiring: Effects of job demands and applicant competence. *Sex Roles*, 9: 307–319.
- *Gomez-Mejia, L. R., & Balkin, D. B. 1984. Faculty satisfaction with pay and other job dimensions under union and nonunion conditions. *Academy of Management Journal*, 27: 591–602.
- *Gomez-Mejia, L. R., & Balkin, D. B. 1992. Determinants of faculty pay: An agency theory perspective. *Academy of Management Journal*, 35: 921–955.
- Gorman, E. H. 2005. Gender stereotypes, same-gender preferences, and organizational variation in the hiring of women: Evidence from law firms. *American Sociological Review*, 70: 702–728.
- Gorman, E. H. 2006. Work uncertainty and the promotion of professional women: The case of law firm partnership. *Social Forces*, 85: 865–890.
- *Gorman, E. H., & Kmec, J. A. 2009. Hierarchical rank and women's organizational mobility: Glass ceilings in corporate law firms. *American Journal of Sociology*, 114: 1428–1474.
- *Graves, L. M., Ruderman, M. N., Ohlott, P. J., & Weber, T. J. 2012. Driven to work and enjoyment of work effects on managers' outcomes. *Journal of Management*, 38: 1655–1680.
- Greckhamer, T., Misangyi, V. F., Elms, H., & Lacey, R. 2008. Using qualitative comparative analysis in management research: An examination of combinations of industry, corporate and business-unit effects. *Organizational Research Methods*, 11: 695–726.
- Greenhaus, J. H., & Parasuraman, S. 1993. Job performance attributions and career advancement prospects: An examination of gender and race effects. *Organizational Behavior and Human Decision Processes*, 55: 273–297.
- *Greenhaus, J. H., Parasuraman, S., & Wormley, W. M. 1990. Effects of race on organizational experiences, job performance evaluations, and career outcomes. *Academy of Management Journal*, 33: 64–86.
- Harbord, R., & Higgins, J. 2008. Meta-regression in STATA. *The Stata Journal*, 8: 492–519.

- Hauser, R. M., & Warren, J. R. 1997. Socioeconomic indexes for occupations: A review, update, and critique. *Sociological Methodology*, 27: 177–298.
- Hedges, L. V., & Olkin, I. 1985. *Statistical methods for meta-analysis*. Orlando, FL: Academic Press.
- Heilman, M. E. 1983. Sex bias in work settings: The lack of fit model. *Research in Organizational Behavior*, 5: 269–298.
- Heilman, M. E. 1994. Affirmative action: Some unintended consequences. *Research in Organizational Behavior*, 16: 125–170.
- *Hekman, D. R., Aquino, K., Owens, B. P., Mitchell, T. R., Schilpzand, P., & Leavitt, K. 2010. An examination of whether and how racial and gender biases influence customer satisfaction. *Academy of Management Journal*, 53: 238–264.
- *Hersch, J., & Viscusi, W. 1996. Gender differences in promotions and wages. *Industrial Relations: A Journal of Economy and Society*, 35: 461–472.
- Higgins, J., & Thompson, S. 2004. Controlling the risk of spurious findings from meta-regression. *Statistics in Medicine*, 23: 1663–1682.
- Hirsch, C. E., & Kornrich, S. 2008. The context of discrimination: Workplace conditions, institutional environments, and sex and race discrimination charges. *American Journal of Sociology*, 113: 1394–1432.
- Hodge, R. W., Siegel, P. M., & Rossi, P. H. 1964. Occupational prestige in the United States, 1925–63. *American Journal of Sociology*, 70: 286–302.
- *Honeycutt, T. L., & Rosen, B. 1997. Family friendly human resource policies, salary levels, and salient identity as predictors of organizational attraction. *Journal of Vocational Behavior*, 50: 271–290.
- *Hoobler, J. M., Lemmon, G., & Wayne, S. J. 2014. Women's managerial aspirations: An organizational development perspective. *Journal of Management*, 40: 703–730.
- *Hoobler, J. M., Wayne, S. J., & Lemmon, G. 2009. Bosses' perceptions of family-work conflict and women's promotability: Glass ceiling effects. *Academy of Management Journal*, 52: 939–957.
- Huffman, M. L., Cohen, P. N., & Pearlman, J. 2010. Engendering change: Organizational dynamics and workplace gender desegregation, 1975–2005. *Administrative Science Quarterly*, 55: 255–277.
- *Hui, C., Lam, S. S., & Law, K. K. 2000. Instrumental values of organizational citizenship behavior for promotion: A field quasi-experiment. *The Journal of Applied Psychology*, 85: 822–828.
- Hultin, M., & Szulkin, R. 2003. Mechanisms of inequality: Unequal access to organizational power and the gender wage gap. *European Sociological Review*, 19: 143–159.
- Ibarra, H. 1992. Homophily and differential returns: Sex differences in network structure and access in an advertising firm. *Administrative Science Quarterly*, 37: 422–447.
- Ibarra, H., Carter, N., & Silva, C. 2010. Why men still get more promotions than women. *Harvard Business Review*, 88: 80–85.
- *Igbaria, M., & Baroudi, J. 1995. The impact of job performance evaluation on career advancement prospects: An examination of gender differences in the workplace. *Management Information Systems Quarterly*, 19: 5–15.
- *Indartono, S., & Chen, C. V. 2010. Moderation of gender on the relationship between task characteristics and performance. *International Journal of Organizational Innovation*, 2: 195–200.
- *Iverson, R. D., & Kuruvilla, S. 1995. Antecedents of union loyalty: The influence of individual dispositions and organizational context. *Journal of Organizational Behavior*, 16: 557–582.
- *Iverson, R. D., & Pullman, J. A. 2000. Determinants of voluntary turnover and layoffs in an environment of repeated downsizing following a merger: An event history analysis. *Journal of Management*, 26: 977–1003.
- *James, E. H. 2000. Race-related differences in promotions and support: Underlying effects of human and social capital. *Organization Science*, 11: 493–508.
- *Jepsen, D. M., & Rodwell, J. J. 2009. Justice in the workplace: The centrality of social versus judgmental predictors of performance varies by gender. *International Journal of Human Resource Management*, 20: 2066–2083.
- Joshi, A. 2014. By whom and when is women's expertise recognized? The interactive effects of gender and education in science and engineering teams. *Administrative Science Quarterly*, 59: 202–239.
- Joshi, A., & Knight, A. 2015. Who defers to whom and why? Dual mechanisms linking demographic differences to dyadic deference and team effectiveness. *Academy of Management Journal*, 58: 59–84.
- *Joshi, A., Liao, H., & Jackson, S. E. 2006. Cross-level effects of workplace diversity on sales performance and pay. *Academy of Management Journal*, 49: 459–481.
- *Judge, T. A., & Cable, D. M. 2011. When it comes to pay, do the thin win? The effect of weight on pay for men and women. *The Journal of Applied Psychology*, 96: 95.
- *Judge, T. A., & Watanabe, S. 1993. Another look at the job satisfaction–life satisfaction relationship. *The Journal of Applied Psychology*, 78: 939–948.

- *Judge, T. A., Cable, D. M., Boudreau, J. W., & Bretz, R. D. 1995. An empirical investigation of the predictors of executive career success. *Personnel Psychology*, 48: 485–519.
- *Judiesch, M. K., & Lyness, K. S. 1999. Left behind? The impact of leaves of absence on managers' career success. *Academy of Management Journal*, 42: 641–651.
- Kalev, A., Dobbin, F., & Kelly, E. 2006. Best practices or best guesses? Assessing the efficacy of corporate affirmative action and diversity policies. *American Sociological Review*, 71: 589–617.
- *Kamdar, D., & Van Dyne, L. 2007. The joint effects of personality and workplace social exchange relationships in predicting task performance and citizenship performance. *The Journal of Applied Psychology*, 92: 1286.
- Kanter, R. M. 1977. *Men and women of the corporation*. New York: Basic Books.
- Kay, F. M., & Hagan, J. 1998. Raising the bar: The gender stratification of law-firm capital. *American Sociological Review*, 64: 728–743.
- *Keller, R. T. 2012. Predicting the performance and innovativeness of scientists and engineers. *The Journal of Applied Psychology*, 97: 225–233.
- Kepes, S., Banks, G. C., McDaniel, M., & Whetzel, D. L. 2012. Publication bias in the organizational sciences. *Organizational Research Methods*, 15: 624–662.
- *Kirchmeyer, C. 1995. Demographic similarity to the work group: A longitudinal study of managers at the early career stage. *Journal of Organizational Behavior*, 16: 67–83.
- *Kirchmeyer, C. 1998. Determinants of managerial career success: Evidence and explanation of male/female differences. *Journal of Management*, 24: 673–692.
- *Kirchmeyer, C. 2002. Gender differences in managerial careers: Yesterday, today, and tomorrow. *Journal of Business Ethics*, 37: 5–24.
- *Kluemper, D. H., DeGroot, T., & Choi, S. 2013. Emotion management ability predicting task performance, citizenship, and deviance. *Journal of Management*, 39: 878–905.
- *Lam, S. S., Ng, T. W., & Feldman, D. C. 2012. The relationship between external job mobility and salary attainment across career stages. *Journal of Vocational Behavior*, 80: 129–136.
- *Landau, J. 1995. The relationship of race and gender to managers' ratings of promotion potential. *Journal of Organizational Behavior*, 16: 391–400.
- Lansberg, I. 1989. Social categorization, entitlement, and justice in organizations: Contextual determinants and cognitive underpinnings. *Human Relations*, 41: 871–899.
- Lewis-Beck, M. S., Bryman, A. E., & Liao, T. F. 2004. *The Sage encyclopedia of social science research methods, vol. 1*. Thousand Oaks, CA: Sage.
- Lipsey, M. W., & Wilson, D. B. 2001. *Practical meta-analysis*. Thousand Oaks, CA: Sage.
- *Liu, J., Hui, C., Lee, C., & Chen, Z. X. 2013. Why do I feel valued and why do I contribute? A relational approach to employee's organization-based self-esteem and job performance. *Journal of Management Studies*, 50: 1018–1040.
- *Liu, Y., Loi, R., & Lam, L. W. 2011. Linking organizational identification and employee performance in teams: The moderating role of team-member exchange. *International Journal of Human Resource Management*, 22: 3187–3201.
- *Lobel, S. A., & St. Clair, L. 1992. Effects of family responsibilities, gender, and career identity salience on performance outcomes. *Academy of Management Journal*, 35: 1057–1069.
- *Loi, R., & Ngo, H. Y. 2009. Work outcomes of relational demography in Chinese vertical dyads. *International Journal of Human Resource Management*, 20: 1704–1719.
- *Lyness, K. S., & Heilman, M. E. 2006. When fit is fundamental: Performance evaluations and promotions of upper-level female and male managers. *The Journal of Applied Psychology*, 91: 777–785.
- *Lyness, K. S., & Judiesch, M. K. 1999. Are women more likely to be hired or promoted into management positions? *Journal of Vocational Behavior*, 54: 158–173.
- *Lyness, K. S., & Judiesch, M. K. 2001. Are female managers quitters? The relationships of gender, promotions, and family leaves of absence to voluntary turnover. *The Journal of Applied Psychology*, 86: 1167–1178.
- *Lyness, K. S., & Thompson, D. E. 1997. Above the glass ceiling? A comparison of matched samples of female and male executives. *The Journal of Applied Psychology*, 82: 359–375.
- Martin, J., & Myerson, D. D. 1998. Women and power: Conformity, resistance, and disorganized coercion. In R. Kramer & M. Neale (Eds.), *Power and influence in organizations*: 311–348. Thousand Oaks, CA: Sage.
- *Martins, L. L., Eddleston, K. A., & Veiga, J. F. 2002. Moderators of the relationship between work-family conflict and career satisfaction. *Academy of Management Journal*, 45: 399–409.
- McCall, L. 2001. *Complex inequality: Gender, class, and race in the new economy*. New York: Routledge.

- McCracken, D. M. 2000. Winning the talent war on women: Sometimes it takes a revolution. *Harvard Business Review*, 78: 159–167.
- McGahan, A. M., & Porter, M. E. 1997. How much does industry matter, really? *Strategic Management Journal*, 18: 15–30.
- *Miller, J. G., & Wheeler, K. G. 1992. Unraveling the mysteries of gender differences in intentions to leave the organization. *Journal of Organizational Behavior*, 13: 465–478.
- Misangyi, V. F., & Acharya, A. 2014. Substitutes or complements? A configurational examination of corporate governance mechanisms. *Academy of Management Journal*, 57: 1681–1705.
- Mobley, W. H. 1982. Supervisor and employee race and sex effects on performance appraisals: A field study of adverse impact and generalizability. *Academy of Management Journal*, 25: 598–606.
- Morrison, A. M., & Von Glinow, M. A. 1990. Women and minorities in management. *The American Psychologist*, 45: 200–208.
- *Morrison, E. W. 1994. Role definitions and organizational citizenship behavior: The importance of the employee's perspective. *Academy of Management Journal*, 37: 1543–1567.
- *Mount, M. K., In-Sue, O., & Burns, M. 2008. Incremental validity of perceptual speed and accuracy over general mental ability. *Personnel Psychology*, 61: 113–139.
- Nakao, K., & Treas, J. 1994. Updating occupational prestige and socioeconomic scores: How the new measures measure up. In P. Marsden (Ed.), *Sociological methodology*: 1–72, Washington, DC: American Sociological Association.
- Nelson, R. R., & Bridges, W. P. 1999. *Legalizing gender inequality: Court, markets, and unequal pay for women in America*. Cambridge: Cambridge University Press.
- Ng, T. W. H., Eby, L. T., Sorensen, K. L., & Feldman, D. C. 2005. Predictors of objective and subjective career success: A meta-analysis. *Personnel Psychology*, 58: 367–408.
- Niederle, M., & Vesterlund, L. 2007. Do women shy away from competition? Do men compete too much? *The Quarterly Journal of Economics*, 122: 1067–1101.
- Nieva, V. F., & Gutek, B. A. 1980. Sex effects on evaluation. *Academy of Management Review*, 5: 267–276.
- *Nyberg, A. J., Pieper, J. R., & Trevor, C. O. 2013. Pay-for-performance's effect on future employee performance integrating psychological and economic principles toward a contingency perspective. *Journal of Management*, Advance online publication. [DOI: 10.1177/0149206313515520].
- Oritz, S., & Roscigno, V. 2009. Discrimination, women, and work: Processes and variations by race and class. *The Sociological Quarterly*, 50: 336–359.
- Ostroff, C., & Atwater, L. E. 2003. Does whom you work with matter? Effects of referent group gender and age composition on managers' compensation. *The Journal of Applied Psychology*, 88: 725–740.
- *Ostroff, C., Atwater, L. E., & Feinberg, B. J. 2004. Understanding self-other agreement: A look at rater and ratee characteristics, context, and outcomes. *Personnel Psychology*, 57: 333–375.
- *Peterson, S. J., Luthans, F., Avolio, B. J., Walumbwa, F. O., & Zhang, Z. 2011. Psychological capital and employee performance: A latent growth modeling approach. *Personnel Psychology*, 64: 427–450.
- Podsakoff, N. P., Whiting, S. W., Podsakoff, P. M., & Blume, B. D. 2009. Individual- and organizational-level consequences of organizational citizenship behaviors: A meta-analysis. *The Journal of Applied Psychology*, 94: 122–141.
- Porter, M. E. 1980. *Competitive strategy*. New York: Free Press.
- *Powell, G. N., & Butterfield, D. A. 1994. Investigating the "glass ceiling" phenomenon: An empirical study of actual promotions to top management. *Academy of Management Journal*, 37: 68–86.
- Pratto, F., Stallworth, L. M., Sidanius, J., & Siers, B. 1997. The gender gap in occupational role attainment: A social dominance approach. *Journal of Personality and Social Psychology*, 72: 37–53.
- Ragin, C. C. 2000. *Fuzzy-set social science*. Chicago, IL: University of Chicago Press.
- Ragin, C. C. 2008. *Redesigning social inquiry: Fuzzy sets and beyond*. Chicago, IL: University of Chicago Press.
- Ragin, C. C., & Rihoux, B. 2009. Qualitative comparative analysis (QCA) and related systematic comparative methods. *International Sociology*, 21: 679–706.
- *Ragins, B. R., & Cornwell, J. M. 2001. Pink triangles: Antecedents and consequences of perceived workplace discrimination against gay and lesbian employees. *The Journal of Applied Psychology*, 86: 1244–1267.
- *Randel, A. E., & Jaussi, K. S. 2003. Functional background identity, diversity, and individual performance in cross-functional teams. *Academy of Management Journal*, 46: 763–774.
- Reskin, B. F. 2003. Including mechanisms of ascriptive inequality. *American Sociological Review*, 68: 1–21.
- *Restubog, S. L. D., Bordia, P., & Tang, R. L. 2007. Behavioural outcomes of psychological contract breach in a non-Western culture: The moderating role of equity sensitivity. *British Journal of Management*, 18: 376–386.

- *Restubog, S. L. D., Bordia, P., Tang, R. L., & Krebs, S. A. 2010. Investigating the moderating effects of leader-member exchange in the psychological contract breach-employee performance relationship: A test of two competing perspectives. *British Journal of Management*, 21: 422-437.
- Ridgeway, C. 1991. The social construction of status value: Gender and other nominal characteristics. *Social Forces*, 70: 367-386.
- *Rosin, H., & Korabik, K. 1995. Organizational experiences and propensity to leave: A multivariate investigation of men and women managers. *Journal of Vocational Behavior*, 46: 1-16.
- Rudman, L. A., Moss-Racusin, C. A., Phelan, J. E., & Nauts, S. 2012. Status incongruity and backlash effects: Defending the gender hierarchy motivates prejudice against female leaders. *Journal of Experimental Social Psychology*, 48: 165-179.
- Sackett, P. R., DuBois, C. L., & Noe, A. W. 1991. Tokenism in performance evaluation: The effects of work group representation on male-female and white-black differences in performance ratings. *The Journal of Applied Psychology*, 76: 263-267.
- *Salamin, A., & Hom, P. W. 2005. In search of the elusive U-shaped performance-turnover relationship: Are high-performing Swiss bankers more liable to quit? *The Journal of Applied Psychology*, 90: 1204-1230.
- *Schaubroeck, J., & Lam, S. S. 2002. How similarity to peers and supervisor influences organizational advancement in different cultures. *Academy of Management Journal*, 45: 1120-1136.
- *Schneer, J. A., & Reitman, F. 1990. Effects of employment gaps on the careers of MBAs: More damaging for men than for women? *Academy of Management Journal*, 33: 391-406.
- *Schneer, J. A., & Reitman, F. 1993. Effects of alternate family structures on managerial career paths. *Academy of Management Journal*, 36: 830-843.
- *Schneer, J. A., & Reitman, F. 1994. The importance of gender in mid-career: A longitudinal study of MBAs. *Journal of Organizational Behavior*, 15: 199-207.
- *Seibert, S. E., Grant, J. M., & Kraimer, M. L. 1999. Proactive personality and career success. *The Journal of Applied Psychology*, 84: 416-427.
- *Seibert, S. E., Kraimer, M. L., & Liden, R. C. 2001. A social capital theory of career success. *Academy of Management Journal*, 44: 219-237.
- *Shaw, J. D., Duffy, M. K., Mitra, A., Lockhart, D. E., & Bowler, M. 2003. Reactions to merit pay increases: A longitudinal test of a signal sensitivity perspective. *The Journal of Applied Psychology*, 88: 538-553.
- *Shi, J., Chen, Z., & Zhou, L. 2011. Testing differential mediation effects of sub-dimensions of political skills in linking proactive personality to employee performance. *Journal of Business and Psychology*, 26: 359-369.
- *Shoss, M. K., Eisenberger, R., Restubog, S. L. D., & Zagenczyk, T. J. 2013. Blaming the organization for abusive supervision: The roles of perceived organizational support and supervisor's organizational embodiment. *The Journal of Applied Psychology*, 98: 158-168.
- Sidanius, J., & Pratto, F. 1999. *Social dominance: An intergroup theory of social hierarchy and oppression*. New York: Cambridge University Press.
- Sidanius, J., Pratto, F., Sinclair, S., & van Laar, C. 1996. Mother Teresa meets Genghis Khan: The dialectics of hierarchy-enhancing and hierarchy-attenuating career choices. *Social Justice Research*, 9: 145-170.
- Siegel, P. M. 1971. *Prestige in the American occupational structure*. Unpublished PhD thesis, University of Chicago, Chicago, IL.
- Sinclair, S., Sidanius, J., & Levin, S. 2010. The interface between ethnic and social system attachment: The differential effects of hierarchy-enhancing and hierarchy-attenuating environments. *The Journal of Social Issues*, 54: 741-757.
- *Sosik, J. J., Juzbasich, J., & Chun, J. U. 2011. Effects of moral reasoning and management level on ratings of charismatic leadership, in-role and extra-role performance of managers: A multi-source examination. *The Leadership Quarterly*, 22: 434-450.
- Speilerman, S., & Petersen, T. 1999. Organizational structure, determinants of promotion, and gender differences in attainment. *Social Science Research*, 28: 203-227.
- Spurr, S. J., & Sueyoshi, G. T. 1994. Turnover and promotion of lawyers: An inquiry into gender differences. *The Journal of Human Resources*, 29: 813-842.
- Stainback, K., Tomaskovic-Devey, D., & Skaggs, S. 2010. Organizational approaches to inequality: Inertia, relative power, and environments. *Annual Review of Sociology*, 36: 225-247.
- Stroh, L. K., Brett, J. M., & Reilly, A. H. 1992. All the right stuff: A comparison of female and male managers' career progression. *The Journal of Applied Psychology*, 77: 251-260.
- *Sturges, J., Guest, D., Conway, N., & Mackenzie Davey, K. 2002. A longitudinal study of the relationship between career management and organizational commitment among graduates in the first ten years at work. *Journal of Organizational Behavior*, 23: 731-748.
- *Sturman, M. C., Shao, L., & Katz, J. H. 2012. The effect of culture on the curvilinear relationship between performance and turnover. *The Journal of Applied Psychology*, 97: 46.

- *Tharenou, P. 2001. Going up? Do traits and informal social processes predict advancing in management? *Academy of Management Journal*, 44: 1005–1017.
- Thompson, S. G., & Higgins, J. P. T. 2002. How should meta-regression analyses be undertaken and interpreted? *Statistics in Medicine*, 21: 1559–1573.
- Tilly, C. 1998. *Durable inequality*. Berkeley, CA: University of California Press.
- Tomaskovic-Devey, D. 1993. *Gender and racial inequality at work: The sources and consequences of job segregation*. Ithaca, NY: ILR Press.
- Tomaskovic-Devey, D., & Skaggs, S. 1999. An establishment-level test of the statistical discrimination hypothesis. *Work and Occupations*, 26: 422–445.
- *Tremblay, M., St-Onge, S., & Toulouse, J. M. 1997. Determinants of salary referents relevance: A field study of managers. *Journal of Business and Psychology*, 11: 463–484.
- *Trevor, C. O. 2001. Interactions among actual ease-of-movement determinants and job satisfaction in the prediction of voluntary turnover. *Academy of Management Journal*, 44: 621–638.
- *Tsui, A. S., & O'Reilly, C. A., III. 1989. Beyond simple demographic effects: The importance of relational demography in superior-subordinate dyads. *Academy of Management Journal*, 32: 402–423.
- *Turban, D. B., & Dougherty, T. W. 1994. Role of protégé personality in receipt of mentoring and career success. *Academy of Management Journal*, 37: 688–702.
- *Turnley, W. H., & Feldman, D. C. 2000. Re-examining the effects of psychological contract violations: Unmet expectations and job dissatisfaction as mediators. *Journal of Organizational Behavior*, 21: 25–42.
- *Turnley, W. H., Bolino, M. C., Lester, S. W., & Bloodgood, J. M. 2003. The impact of psychological contract fulfillment on the performance of in-role and organizational citizenship behaviors. *Journal of Management*, 29: 187–206.
- *Van Dyne, L., & LePine, J. A. 1998. Helping and voice extra-role behaviors: Evidence of construct and predictive validity. *Academy of Management Journal*, 41: 108–119.
- *Van Iddekinge, C. H., Morgeson, F. P., Schleicher, D. J., & Campion, M. A. 2011. Can I retake it? Exploring subgroup differences and criterion-related validity in promotion retesting. *The Journal of Applied Psychology*, 96: 941.
- Vescio, T. K., Gervais, S., Snyder, M., & Hoover, A. 2005. Power and the creation of patronizing environments: The stereotype-based behaviors of the powerful and their effects on female performance in masculine domains. *Journal of Personality and Social Psychology*, 88: 658–672.
- Viswesvaran, C., & Ones, D. S. 1995. Theory testing: Combining psychometric meta-analysis and structural equations modeling. *Personnel Psychology*, 48: 865–886.
- *Wang, M. O., Zhan, Y., Mccune, E., & Truxillo, D. 2011. Understanding newcomers' adaptability and work-related outcomes: Testing the mediating roles of perceived P-E fit variables. *Personnel Psychology*, 64: 163–189.
- *Watkins, M. B., et al. 2006. Does it pay to be a sexist? The relationship between modern sexism and career outcomes. *Journal of Vocational Behavior*, 69: 524–537.
- Wayne, S. J., & Liden, R. C. 1995. Effects of impression management on performance ratings: A longitudinal study. *Academy of Management Journal*, 38: 232–260.
- Weeden, K. A. 2002. Why some occupations pay more than others? Social closure and earnings inequality in United States. *American Journal of Sociology*, 108: 55–101.
- *Weng, Q., & McElroy, J. C. 2012. Organizational career growth, affective occupational commitment and turnover intentions. *Journal of Vocational Behavior*, 80: 256–265.
- *Whitely, W. T., & Coetsier, P. 1993. The relationship of career mentoring to early career outcomes. *Organization Studies*, 14: 419–441.
- *Whitely, W. T., Dougherty, T. W., & Dreher, G. F. 1991. Relationship of career mentoring and socioeconomic origin to managers' and professionals' early career progress. *Academy of Management Journal*, 34: 331–351.
- Whitener, E. M. 1990. Confusion of confidence intervals and credibility intervals in meta-analysis. *The Journal of Applied Psychology*, 75: 315–321.
- Williams, L. J., & Anderson, S. E. 1991. Job satisfaction and organizational commitment as predictors of organizational citizenship and in-role behaviors. *Journal of Management*, 17: 601–617.
- *Witt, L. A., Kacmar, K. M., & Andrews, M. C. 2001. The interactive effects of procedural justice and exchange ideology on supervisor-rated commitment. *Journal of Organizational Behavior*, 22: 505–515.
- *Wu, P. C., Foo, M. D., & Turban, D. B. 2008. The role of personality in relationship closeness, developer assistance, and career success. *Journal of Vocational Behavior*, 73: 440–448.
- *Yao, X., & Wang, L. 2008. Socially oriented values and reciprocity norm predict organizational commitment. *Asian Journal of Social Psychology*, 11: 247–252.

- *Zatzick, C. D., Elvira, M. M., & Cohen, L. E. 2003. When is more better? The effects of racial composition on voluntary turnover. *Organization Science*, 14: 483–496.
- *Zhang, X., & Bartol, K. M. 2010. The influence of creative process engagement on employee creative performance and overall job performance: A curvilinear assessment. *The Journal of Applied Psychology*, 95: 862–873.
- *Zhang, Z., Waldman, D. A., & Wang, Z. 2012. A multi-level investigation of leader–member exchange, informal leader emergence, and individual and team performance. *Personnel Psychology*, 65: 49–78.



Aparna Joshi (aparnajo@psu.edu) is a professor of management at Smeal College of Business, Pennsylvania State University. She received her PhD from the School of Management and Labor Relations, Rutgers University.

She studies various topics in organizational behavior, including gender, diversity, global dispersion in teams, and generational differences in organizations.

Hyuntak Roh (hroh@yonsei.ac.kr) is an associate professor of management at Yonsei University, Seoul, Korea. He received his PhD from University of Illinois, Urbana-Champaign. His research interests include diversity and inequality issues in the workplace, diversity management practices, and cross-cultural/functional team management.

Jooyeon Son (son22@illinois.edu) received her PhD from the School of Labor and Employment Relations at University of Illinois, Urbana-Champaign, in 2015 and is currently a visiting lecturer there. She conducts research in the areas of gender, team diversity, leader–member exchange, and talent management.



Copyright of Academy of Management Journal is the property of Academy of Management and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.